NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Paediatric Ortho

1. DDH
2. Perthes and SCFE
3. clubfoot and foot deformities
4. paediatric fractures
5. cerebral palsy and neuromuscular
6. skeletal dysplasias
7. bone and joint infection in children

DDH

1. Definition and epidemiology

- **Developmental dysplasia of the hip (DDH):** spectrum from **acetabular dysplasia** (shallow acetabulum) through **subluxation** to **frank dislocation** (dislocatable or dislocated), developing **before or after birth** (hence "developmental", not "congenital").
- **Incidence: instability about 1-2 per 100 newborns at birth** (most resolve), **true dislocation about 1-1.5 per 1,000**.
- **Teratologic dislocation** (separate entity): dislocated **in utero**, **irreducible**, with **neuromuscular syndromes** (**arthrogryposis, myelomeningocele, Larsen syndrome**): worse prognosis, early open reduction.

2. Risk factors ("4 Fs" + others)

- **Female (about 4-6:1; maternal relaxin sensitivity)**, **First-born** (tight uterus), **Family history (about 10-12x risk with an affected sibling or parent)**, **breech presentation ("Frank breech": hips flexed, knees extended: strongest single risk factor)**.
- **Oligohydramnios**, **left hip (about 60%; left occiput anterior position against the maternal sacrum)**, bilateral in about 20%.
- **"Packaging" disorders** (associated conditions from intrauterine crowding): **congenital muscular torticollis**, **metatarsus adductus**, **congenital knee dislocation**, **calcaneovalgus/clubfoot** (association weaker).
- **Postnatal swaddling with hips extended and adducted** (Native American, Japanese cultural practices; **"hip-healthy swaddling" with hips free to flex and abduct**).
- **Ethnicity:** higher in Native Americans and Sami; low in Africans (carrying on the back with hips abducted).

3. Pathoanatomy

- **Obstacles to reduction (acetabular side, in order):**
  - **Intra-articular: inverted/hypertrophied labrum ("limbus")**, **hypertrophied ligamentum teres**, **pulvinar (fibrofatty tissue in the acetabular floor)**, **contracted inferior capsule**, **transverse acetabular ligament** (hypertrophied).
  - **Extra-articular: iliopsoas tendon** (creates an **hourglass constriction of the capsule**), **adductor longus** (contracture).
- **Secondary changes: shallow, anteverted acetabulum**, **increased femoral anteversion**, **coxa valga**, **false acetabulum (in neglected cases)**, **neolimbus** formation.

4. Clinical examination by age

Newborn (up to about 3 months)

- **Barlow test** (**adduction + posterior pressure on the flexed hip: dislocatable hip**: **"Barlow = Bye-bye"**).
- **Ortolani test** (**abduction + elevation of the greater trochanter: reduces a dislocated hip with a "clunk"**: **"Ortolani = Out to In"**).
- **High-pitched clicks** (soft tissue) are not significant.
- **Both tests become negative after about 3 months** as soft tissues tighten (dislocated hips become irreducible and fixed).

Older infant (over 3 months)

- **Limited abduction (most reliable sign: under 60 degrees in flexion or asymmetry)**.
- **Galeazzi (Allis) sign**: **apparent femoral shortening** with hips and knees flexed (unilateral dislocation; **negative in bilateral DDH**).
- **Asymmetric thigh/gluteal skin folds** (unreliable), **Klisic test** (line from the ASIS through the greater trochanter passes below the umbilicus in a dislocated hip).

Walking child

- **Trendelenburg gait**, **limp**, **leg-length discrepancy**, **waddling gait** (bilateral), **hyperlordosis** (bilateral).

5. Imaging

Ultrasound (up to about 4-6 months: before the femoral head ossific nucleus appears)

- **Graf method (static, coronal view):**
  - **Alpha angle** (**bony roof**: angle between the baseline (ilium) and the bony acetabular roof line): **normal 60 degrees or more**.
  - **Beta angle** (**cartilaginous roof**: baseline to the labrum/cartilage roof line): **normal under 55 degrees** (increases as the head displaces).

Graf type	Alpha	Beta	Description	Management
I	**≥60**	<55	**Normal (mature)**	None
IIa	50-59	55-77	**Physiological immaturity (under 3 months)**	**Repeat ultrasound at 6 weeks to 3 months**
IIb	50-59		Delayed ossification (**over 3 months**)	**Pavlik harness**
IIc	**43-49**	<77	**Critical** (stable or unstable)	**Pavlik**
D	43-49	**>77**	**Decentring** (about to dislocate)	**Pavlik**
III	**<43**	>77	**Dislocated (cartilage roof pushed up)**	Pavlik/closed reduction

Graf type	Alpha	Beta	Description	Management
IV	<43	Labrum inverted (pushed down)	**Dislocated (labrum inverted)**	Closed/open reduction

- **Dynamic ultrasound (Harcke)**: stability; **femoral head coverage (Terjesen/Morin): under 50% abnormal**.
- **Selective ultrasound screening** (**UK NIPE**: risk factors (**breech at or after 36 weeks, family history**) or abnormal examination: **ultrasound by 6 weeks**) vs **universal ultrasound screening** (Austria, Germany: more treatment, possibly fewer late cases).

Radiographs (after about 4-6 months, once the ossific nucleus appears)

- **Hilgenreiner line** (horizontal through the triradiate cartilages), **Perkin line** (vertical from the lateral acetabular margin): **normal: femoral head ossific nucleus (or medial metaphyseal beak) in the inferomedial quadrant**.
- **Shenton line** (broken in subluxation/dislocation).
- **Acetabular index** (**normal under about 30 degrees at birth, decreasing to under 20-25 degrees by 2 years**): increased in dysplasia.
- **Delayed appearance of the ossific nucleus** on the affected side.
- **Centre-edge angle of Wiberg (after about 5 years): normal over 20-25 degrees**.
- **Arthrography** (at the time of closed reduction): **medial dye pool (under 5 mm = acceptable concentric reduction)**, inverted limbus, **"rose thorn" sign** (normal labrum).
- **MRI/CT after reduction** in spica: confirm reduction (**MRI preferred: no radiation**).

6. Treatment by age

0-6 months: Pavlik harness

- **Dynamic flexion-abduction orthosis**: **hips flexed about 100 degrees** (anterior straps), **abduction by gravity in the "safe zone" (Ramsey)** (posterior straps loose: prevent adduction but do not force abduction).
- **Safe zone (Ramsey):** the arc between **maximum abduction** and the **position of redislocation (adduction)**: **avoid forced abduction over about 60 degrees** (**AVN**).
- **Positioning errors:**
  - **Excessive flexion (over 120 degrees)**: **femoral nerve palsy** (commonest nerve complication; quadriceps weakness: **stop/adjust harness**), **inferior (obturator) dislocation**,
  - **Insufficient flexion**: failure to reduce,
  - **Forced abduction**: **AVN of the femoral head**,
  - **Brachial plexus palsy** (shoulder straps too tight), skin breakdown.
- **Monitor with ultrasound**: **if not reduced by 3-4 weeks, abandon the Pavlik** (**"Pavlik harness disease"**: **erosion of the posterior acetabular wall** from prolonged use with a persistently dislocated (posterior) head).
- **Success: about 90%+ for dysplastic/subluxated hips**, **about 80-85% for dislocated (Ortolani-positive) hips**; **less effective with Graf IV, bilateral dislocations, Barlow-negative irreducible hips, older age (over 7 weeks at initiation), absent Ortolani sign**.
- **Duration:** full-time until stable (usually about **6-12 weeks**) then weaning.

6-18 months: closed reduction + hip spica

- **Under general anaesthesia**, **arthrogram**, **percutaneous adductor tenotomy** (increases the safe zone).

- **Hip spica cast in "human position" (Salter): hip flexion about 100 degrees, abduction about 40-50 degrees (avoid over 60), neutral rotation**: **about 3 months** in spica (change at 6 weeks), then abduction brace.
- **Post-reduction imaging: MRI or CT** (concentricity).
- **Traction before reduction** (historical: to reduce AVN): **not supported by evidence**.
- **Ossific nucleus presence:** does not affect AVN risk significantly in most studies; **delaying reduction until it appears is not recommended**.

Open reduction: indications

- **Failed closed reduction** (**medial dye pool over 5-7 mm**, unstable in the safe zone, **excessive abduction needed**), **age over about 18-24 months** (variable), **teratologic** dislocation.
- **Approaches:**
  - **Medial approach (Ludloff: between adductor brevis and pectineus/iliopsoas; or Ferguson: posterior to adductor brevis): under about 12-18 months**; addresses medial obstacles (**iliopsoas, ligamentum teres, transverse acetabular ligament, inferior capsule**); **cannot perform capsulorrhaphy**; **medial femoral circumflex artery at risk: higher AVN**.
  - **Anterior (anterolateral, Smith-Petersen/bikini) approach**: **over about 12-18 months**; allows **capsulorrhaphy** and **concomitant pelvic osteotomy**; **iliopsoas tenotomy**, **inverted labrum**, pulvinar.

Over 18-24 months to 8 years: open reduction + femoral and/or pelvic osteotomy

- **Femoral shortening osteotomy** (**reduces pressure on the head: lowers AVN and redislocation**) + **varus derotation** if needed.
- **Pelvic osteotomies:**

- **Redirectional (reshaping the acetabulum: requires a concentrically reduced, congruous hip)**:
- **Salter innominate osteotomy** (**single cut above the acetabulum from the sciatic notch to the AIIS; hinges on the pubic symphysis**: **anterolateral coverage**; **age 18 months to about 6 years**; **acetabular index correction up to about 10-15 degrees**; open triradiate cartilage),
- **Triple innominate osteotomy (Steel, Tönnis)** (**older children/adolescents**; ilium, ischium, pubis),
- **Periacetabular osteotomy (Bernese/Ganz PAO)** (**closed triradiate cartilage: adolescents and adults**; **posterior column preserved**: allows early weight bearing, preserves pelvic ring),
- **Reshaping (volume-reducing: incomplete osteotomies hinging on the triradiate cartilage; for capacious acetabula; open triradiate cartilage):**
- **Pemberton (pericapsular)** (**hinges on the triradiate cartilage**; **reduces acetabular volume**; **anterolateral coverage**; **age 18 months to 10 years**),
- **Dega** (hinges on the triradiate cartilage; **posterolateral coverage**: **neuromuscular dysplasia (cerebral palsy: posterosuperior deficiency)**),
- **San Diego** (neuromuscular).
- **Salvage (augmenting with capsular interposition: fibrocartilage metaplasia; for incongruent hips):**
- **Chiari osteotomy** (**medial displacement of the acetabulum beneath the ilium**: **capsule interposed**),
- **Shelf procedure (Staheli)** (bone graft extending the lateral acetabulum).

Over 8 years (bilateral) or over about 10 years (unilateral)

- **Reduction often not attempted** for **high bilateral dislocations without a false acetabulum** (**better function than with**

AVN/failed reduction); unilateral: still possible in selected cases; **salvage/arthroplasty later**.

7. Complications

- **Avascular necrosis (AVN) of the femoral head**: **the most important iatrogenic complication**; caused by **forced abduction, excessive pressure, medial circumflex artery injury**: **Kalamchi and MacEwen classification**:
  - **Group I: changes in the ossific nucleus only** (best prognosis),
  - **Group II: lateral physeal damage** (**valgus tilt** of the head, **lateral growth arrest**),
  - **Group III: central physeal damage** (short neck, **coxa breva**, trochanteric overgrowth),
  - **Group IV: total head and physeal damage** (worst).
  - (**Bucholz-Ogden classification** also used.)
- **Redislocation**, **residual dysplasia** (needs pelvic osteotomy), **femoral nerve palsy** (Pavlik), **Pavlik disease**, **stiffness**, **early OA** (**the commonest long-term sequela: the leading cause of THA under age 40-50**), **leg-length discrepancy**, **trochanteric overgrowth** (**greater trochanteric epiphysiodesis or advancement**).
- **Residual dysplasia** is followed with serial radiographs to skeletal maturity.

8. Adult sequelae (DDH in adults)

- **Crowe classification (proximal migration of the femoral head)**: **I under 50% subluxation**, **II 50-75%**, **III 75-100%**, **IV over 100% (high dislocation)**.
- **Hartofilakidis: dysplasia**, **low dislocation** (false acetabulum overlapping the true), **high dislocation** (no contact).

- **THA in DDH: cup in the true acetabulum** (low hip centre), **small cups**, **structural autograft (femoral head) for superolateral deficiency**, **subtrochanteric shortening osteotomy (Crowe IV)** (lengthening **over about 4 cm: sciatic nerve palsy risk**), **excessive femoral anteversion (modular/cemented stems)**.

Exam pearl: **Barlow = dislocatable (adduct + push back); Ortolani = dislocated reduces (abduct + lift). After 3 months: limited abduction is the best sign; Galeazzi negative if bilateral. Breech + female + first-born + family history.**

Exam pearl: **Graf: alpha ≥60 = type I; IIa physiological under 3 months; under 43 = dislocated (III/IV). Ultrasound under 4-6 months, radiographs after (Hilgenreiner, Perkin, Shenton, acetabular index).**

Exam pearl: **0-6 months Pavlik (flexion about 100, no forced abduction; excess flexion = femoral nerve palsy; stop at 3-4 weeks if not reduced: Pavlik disease). 6-18 months closed reduction + spica (human position, adductor tenotomy, medial dye pool under 5 mm). Over 18-24 months open reduction + femoral shortening ± pelvic osteotomy. AVN (Kalamchi-MacEwen) from forced abduction.**

Exam pearl: **Salter: hinge at symphysis, open triradiate, congruent hip. Pemberton/Dega: hinge at the triradiate (Dega: CP, posterior coverage). PAO: closed triradiate, posterior column intact. Chiari/shelf: salvage for incongruent hips (capsular metaplasia).**

Perthes and SCFE

Part A: Legg-Calvé-Perthes disease (LCPD)

1. Definition and epidemiology

- **Idiopathic avascular necrosis of the capital femoral epiphysis in a child**, followed by **fragmentation, revascularisation and remodelling** (self-limiting but may leave a deformed head).
- **Age 4-8 years (range 2-12)**; **boys about 4-5:1**; **bilateral in about 10-15%** (**asynchronous**: **symmetric simultaneous bilateral involvement suggests an alternative diagnosis: multiple epiphyseal dysplasia (MED), hypothyroidism, sickle cell, Gaucher, spondyloepiphyseal dysplasia**).
- **Associations: delayed bone age (about 90%)**, **short stature**, **low birth weight**, **ADHD/hyperactivity**, **passive smoking**, **low socioeconomic status**, **thrombophilia (protein C/S deficiency, factor V Leiden: controversial)**, **inguinal hernia**, genitourinary anomalies.

2. Pathogenesis

- **Single or repeated interruption of the blood supply** (**lateral epiphyseal vessels from the medial femoral circumflex artery**) to the capital femoral epiphysis (**at age 4-7, the epiphysis depends almost entirely on the lateral epiphyseal vessels**: the **ligamentum teres artery contributes little** and the **metaphyseal vessels do not cross the physis**).
- **Stages (Waldenström):**
  - **Initial (sclerosis/necrosis)**: **sclerotic, smaller ossific nucleus**, **widened medial joint space**, **subchondral fracture ("crescent sign": Caffey sign)**: about 6 months,

- **Fragmentation (resorption)**: **fragmented epiphysis**, lateral extrusion, **most mechanically vulnerable stage** (deformation): about 1 year,
- **Reossification (healing)**: new bone formation: about 3-5 years,
- **Residual (healed)**: final shape (**coxa magna, coxa plana, short neck, trochanteric overgrowth**).

3. Clinical features

- **Painless (or mild) limp** (**commonest presentation**), **pain referred to the knee/thigh** (obturator nerve), **antalgic or Trendelenburg gait**, **limited abduction and internal rotation** (most important; **loss of abduction = poor prognosis**), thigh atrophy, mild shortening.
- **Differential diagnosis: transient synovitis**, **septic arthritis**, **MED/SED**, **hypothyroidism**, **sickle cell disease**, **Gaucher disease**, **steroid-induced AVN**, **Meyer dysplasia** (**irregular ossification of the femoral head under 4 years**, often bilateral: **self-resolving**).

4. Imaging

- **Radiographs (AP and frog-leg lateral)**: stages, **crescent sign (subchondral fracture: earliest radiographic sign)**, head-at-risk signs.
- **MRI** (early diagnosis, extent of necrosis, **perfusion MRI**), **bone scan** (avascularity; early).
- **Arthrography** (dynamic: **hinge abduction**, containability).

5. Classifications

Herring lateral pillar classification (**most widely used; prognosis; assessed at the start of the fragmentation stage**)

- **Lateral pillar** = lateral 15-30% of the epiphysis on AP radiograph.

Group	Lateral pillar height	Prognosis
A	**Intact (100%)**	**Good regardless of age**
B	**Over 50% maintained**	Good if **under 8 years at onset** (bone age under 6)
B/C border	**About 50%**, narrow (2-3 mm) or poorly ossified pillar, or exactly 50%	Intermediate
C	**Under 50% maintained**	**Poor** (regardless of treatment, especially older)

Catterall classification (extent of head involvement: I-IV; interobserver reliability poor)

- **I: anterior head only (under 25%)**; **II: under 50% (anterior + central sequestrum)**; **III: over 50% (only medial/posterior spared)**; **IV: whole head**.
- **Catterall "head-at-risk" signs (poor prognosis):**
  - **Gage sign** (V-shaped radiolucency in the lateral epiphysis/metaphysis),
  - **Calcification lateral to the epiphysis** (lateral extrusion),
  - **Lateral subluxation of the femoral head**,
  - **Horizontal growth plate**,
  - **Metaphyseal cysts (diffuse metaphyseal reaction)**.

Salter-Thompson classification

- **A: subchondral fracture under 50% of the head; B: over 50%** (prognostic; in the early stage).

Stulberg classification (outcome at maturity: predicts OA risk)

Class	Head shape	Congruency	OA risk
I	**Normal (spherical)**	**Spherical congruency**	**None**

Class	Head shape	Congruency	OA risk
II	**Spherical but larger (coxa magna), short neck or steep acetabulum**	**Spherical congruency**	Minimal
III	**Ovoid/mushroom-shaped (not flat)**	**Aspherical congruency**	**Mild-moderate OA in later adulthood**
IV	**Flat head (over 1 cm flat weight-bearing surface) + flat acetabulum**	**Aspherical congruency**	**Moderate OA (in middle age)**
V	**Flat head with normal acetabulum**	**Aspherical incongruency**	**Severe early OA**

6. Prognostic factors

- **Age at onset (most important): under 6 years: good prognosis** (more remodelling time); **over 8 years: poor**.
- **Lateral pillar classification (Herring)** (most important radiographic predictor), **extent of head involvement**, **head-at-risk signs**, **female sex** (worse: earlier skeletal maturity), **loss of hip motion (abduction)**, **hinge abduction**, **Stulberg class at maturity**.

7. Management (principle: containment)

- **Goals: maintain the femoral head within the acetabulum (containment)** during the vulnerable fragmentation stage, so that the acetabulum acts as a mould; **maintain ROM** (especially **abduction**).
- **Non-operative:**
  - **Under 6 years (bone age), all groups (mostly): observation, activity modification, physiotherapy (ROM, abduction stretching)**, NSAIDs for synovitis; **bracing (Petrie**

casts/abduction orthoses): largely abandoned (no proven benefit).

- **Operative containment (Herring prospective study: 2004):**
  - **Over 8 years (chronological; bone age over 6) with lateral pillar B or B/C**: **surgical containment (proximal femoral varus osteotomy or Salter innominate osteotomy) gives better results than non-operative treatment**.
  - **Group A**: good outcome regardless.
  - **Group C**: poor outcome regardless of treatment (in most series).
  - **Under 8 years with B**: good outcomes without surgery.
- **Containment procedures:**
  - **Proximal femoral varus (derotation) osteotomy (VDRO)**: **about 15-20 degrees varus**; **shortening, Trendelenburg gait (abductor weakness), trochanteric overgrowth** (often remodels in young children),
  - **Salter innominate osteotomy** (**anterolateral coverage**; lengthens slightly; **requires near-full abduction**),
  - **Shelf acetabuloplasty**, **triple innominate osteotomy**, **combined femoral and pelvic osteotomies** for severe cases.
- **Hinge abduction (salvage: non-containable head; the extruded head hinges on the lateral acetabular edge in abduction):**
  - **Valgus (extension) osteotomy** (**brings the congruent part of the head into the weight-bearing zone; improves adduction-abduction mechanics**), **Chiari osteotomy**, **shelf procedure**, **cheilectomy** (removing the lateral prominence: risk of AVN/physeal injury).
- **Late sequelae:**
  - **Trochanteric overgrowth (relative)**: **greater trochanteric epiphysiodesis (under 8 years)** or **trochanteric advancement (Wagner)**.

- **FAI (cam-type from a mushroom head)**: **surgical hip dislocation with femoral head reduction osteotomy**, **relative femoral neck lengthening**, osteochondroplasty.
- **Osteochondritis dissecans of the femoral head**, **labral tears**, **OA (THA in adulthood)**.

Part B: Slipped capital femoral epiphysis (SCFE)

8. Definition and epidemiology

- **Displacement of the proximal femoral metaphysis anteriorly and superiorly (externally rotated) relative to the epiphysis** (the **epiphysis stays in the acetabulum** and is **posterior and inferior** relative to the neck) through the **hypertrophic zone of the physis**.
- **Commonest adolescent hip disorder**: **age 10-16 (boys: about 13-14; girls: about 11-12)**, **boys more than girls (about 1.5-2:1)**, **left more than right**, **bilateral in about 20-40% (up to 60% eventually; about 20% at presentation)**.
- **Obesity (strongest association: over 95th percentile BMI in about 50-80%)**, **African-American and Polynesian ethnicity**.
- **Endocrine/metabolic causes (atypical SCFE: suspect if under 10 years or over 16, below 50th percentile weight, short stature): hypothyroidism (commonest endocrine)**, **growth hormone deficiency/supplementation**, **renal osteodystrophy (chronic renal failure)**, **hypogonadism**, **panhypopituitarism**, **radiotherapy**, **Down syndrome**. **Bilateral slips are much more common in endocrine SCFE** (**prophylactic pinning recommended**).
- **Biomechanics: increased shear forces** (obesity, **femoral retroversion**, **physeal obliquity: more vertical physis in adolescence**), **perichondrial ring thinning** during growth spurt, **hormonal (oestrogen protects; testosterone weakens the physis)**.

9. Classification

Loder (stability; prognostic: **the most important**)

- **Stable: able to bear weight (with or without crutches)**: **AVN risk about 0-5%**.
- **Unstable: unable to bear weight (even with crutches)**: **AVN risk about 20-50%** (Loder: 47%).

Temporal (traditional)

- **Acute (symptoms under 3 weeks)**, **chronic (over 3 weeks: commonest; remodelling of the metaphysis, posteroinferior callus)**, **acute-on-chronic**.

Southwick slip angle (severity; on frog-leg lateral)

- **Epiphyseal-shaft angle difference vs the contralateral normal side: mild under 30 degrees**, **moderate 30-50 degrees**, **severe over 50 degrees**.

Wilson (displacement on AP)

- **Mild under 33%**, **moderate 33-50%**, **severe over 50%** of the metaphyseal width.

10. Clinical features

- **Groin, thigh or KNEE pain** (**obturator nerve referred pain: knee pain in an adolescent = examine the hip**: **delayed diagnosis common**), **limp**, **out-toeing**.
- **Obligatory external rotation with hip flexion (Drehmann sign)**, **loss of internal rotation** (most sensitive), loss of abduction and flexion, **shortening**.
- **Unstable:** acute severe pain, **unable to bear weight**, limb externally rotated and shortened (like a femoral neck fracture).

11. Imaging

- **AP and frog-leg lateral pelvis** (**frog-leg lateral avoided in unstable slips**: **cross-table lateral instead**: may displace further).
- **Signs:**
  - **Klein line** (**line along the superior femoral neck on AP should intersect the lateral epiphysis; in SCFE it does not or intersects less than the normal side**: **Trethowan sign**),
  - **Widened, irregular physis** (early pre-slip),
  - **Metaphyseal blanch sign of Steel** (**double density: overlap of the displaced epiphysis on the metaphysis**),
  - **Reduced epiphyseal height**,
  - **Capener sign** (loss of the normal triangular overlap of the medial metaphysis with the posterior acetabular wall).
- **MRI: pre-slip (physeal widening, marrow oedema)**; the contralateral hip; AVN.
- **Bone age:** Oxford, **triradiate cartilage open = higher contralateral slip risk**; **Risser**; **modified Oxford score**.

12. Management

Stable SCFE

- **Urgent (not emergency) admission, non-weight bearing**, then **in situ fixation with a single cannulated screw** (**"pin in situ": do NOT attempt reduction: AVN risk**):
  - **Screw perpendicular to the physis, in the centre of the epiphysis on both views**, entering **anterior on the femoral neck** (because the epiphysis is posterior: **anterior-to-posterior trajectory**),
  - **At least 5 threads (about 4-5 threads) across the physis**; **tip at least 5 mm from the subchondral bone** (**avoid joint**

penetration: chondrolysis: **"approach-withdraw" technique** and multiple fluoroscopic views to exclude penetration),
    - **Single central screw for stable slips** (**two screws for unstable slips** add rotational stability but increase penetration risk).
- **Physeal closure** after fixation usually within about 1 year.

Unstable SCFE

- **Emergency (controversial timing): gentle positional/incidental reduction**, **capsular decompression (arthrotomy/aspiration: reduces intracapsular tamponade)**, **fixation with one or two screws**; **urgent fixation within 24 hours** (some evidence for lower AVN); **avoid forceful manipulation**.
- **Modified Dunn procedure (subcapital realignment via surgical hip dislocation: Ganz approach)**: **anatomic reduction with preservation of the retinacular vessels** (**extended retinacular flap**); for **unstable and severe stable slips** in experienced centres; **AVN about 0-25% (debated)**; **intraoperative monitoring of head perfusion (drill hole bleeding, ICP monitoring)**.

Severe chronic/stable slips with deformity

- **In situ pinning followed by corrective osteotomy for symptomatic FAI/deformity later:**
    - **Subcapital (cuneiform: Fish/Dunn)** (**closest to the deformity: best correction, highest AVN**),
    - **Base of neck (Kramer, Barmada)** (intermediate),
    - **Intertrochanteric (Imhäuser, Southwick: flexion-valgus-internal rotation osteotomy)** (**safest for the blood supply, compensatory, less correction**),
    - **Arthroscopic/open osteochondroplasty** for cam deformity.

Prophylactic pinning of the contralateral hip

- **Recommended for: endocrine/renal SCFE**, **age under 10 years**, **open triradiate cartilage**, **poor follow-up compliance**, **modified**

Oxford bone age score under 16-19, **posterior sloping angle (PSA) over about 12-14 degrees** (**increased risk of contralateral slip**). Otherwise: close follow-up.

13. Complications

- **AVN** (**unstable slips (up to about 50%)**, **forceful reduction**, **superior/posterosuperior screw placement** (lateral epiphyseal vessels), **subcapital osteotomy**): **the most serious complication**.
- **Chondrolysis** (**acute cartilage necrosis: joint space narrowing to under 3 mm**, painful stiffness): **unrecognised screw penetration (commonest cause)**, spica casting, **African-American/female**; **ROM physio, NSAIDs**; may need arthrodesis/THA.
- **Slip progression** (fixation failure), **contralateral slip**, **FAI (cam; pistol grip)**, **OA**, **leg-length discrepancy**, **subtrochanteric fracture** (**entry at or below the lesser trochanter** with multiple drill holes: stress riser), infection.

Exam pearl: **Perthes: 4-8-year-old boys, delayed bone age, painless limp, loss of abduction/IR; bilateral symmetrical = think MED/hypothyroidism. Waldenström stages; crescent sign earliest. Herring lateral pillar (A/B/B-C/C) assessed in fragmentation; Catterall head-at-risk signs (Gage, lateral calcification, subluxation, horizontal physis, metaphyseal cysts).**

Exam pearl: **Age at onset is the key prognostic factor (under 6 good). Over 8 years with B or B/C: surgical containment (femoral varus or Salter osteotomy). Hinge abduction: valgus osteotomy. Stulberg I-II good, III-IV moderate, V severe early OA (aspherical incongruity).**

Exam pearl: **SCFE: obese adolescent with knee pain; Klein line, Steel blanch sign; Loder stable (AVN under 5%) vs unstable (up to 50%). Pin in situ with one central screw perpendicular to the physis, no reduction; joint penetration = chondrolysis. Atypical (under 10, thin, endocrine/renal) = bilateral: pin the other side. Imhäuser (intertrochanteric) safest osteotomy; modified Dunn for severe/unstable in expert hands.**

Clubfoot and foot deformities

Part A: Congenital talipes equinovarus (CTEV, clubfoot)

1. Epidemiology and aetiology

- **About 1-2 per 1,000 live births** (higher in **Polynesians (about 7 per 1,000)**, lower in Chinese); **boys 2:1**; **bilateral in about 50%**.
- **Idiopathic (most)**: multifactorial; **genetic: PITX1-TBX4 pathway**, **family history (about 25% risk if a sibling and a parent are affected)**, **maternal smoking**, **early amniocentesis**; **oligohydramnios**.
- **Syndromic/non-idiopathic (stiffer, more recurrence)**: **arthrogryposis**, **myelomeningocele**, **diastrophic dysplasia** ("hitchhiker's thumb", **cauliflower ears**), **Larsen syndrome**, **Freeman-Sheldon**, **Möbius**, **amniotic band syndrome**, **tibial hemimelia**, **Pierre Robin**.
- **Associated: DDH** (examine the hips), **torticollis**.
- **Positional clubfoot** (intrauterine crowding): **flexible, fully correctable passively**: stretching only.

2. Pathoanatomy ("CAVE")

- **C**avus (**midfoot**: **forefoot pronated relative to the hindfoot**: **first ray plantarflexed**),
- **A**dductus (**forefoot adduction**: **medial subluxation of the navicular on the talar head**),
- **V**arus (**hindfoot**: **calcaneus inverted beneath the talus**),
- **E**quinus (**hindfoot/ankle**: **tight Achilles tendon**, plantarflexed talus).

- **Talus: neck medially and plantarly deviated (most deformed bone)**, small, plantarflexed; **navicular medially displaced, articulating with the medial side of the talar head (can reach the medial malleolus)**.
- **Calcaneus:** inverted and adducted, **rotated medially beneath the talus around the interosseous ligament** (**"calcaneus rotates under the talus"**).
- **Soft tissues: contracted posteromedial tissues** (**tibialis posterior, FHL, FDL, Achilles**, deltoid, spring ligament, **master knot of Henry**), **tight plantar fascia**; **calf atrophy (permanent, smaller foot and calf)**; **muscle fibrosis** (increased type I fibres and collagen).
- **Vascular: anterior tibial artery hypoplasia/absence** (about 85% in some studies): **caution with posterior tibial artery during surgery** (dorsalis pedis often absent).

3. Assessment

- **Pirani score** (6 signs, each 0, 0.5, 1; **max 6**):
  - **Midfoot contracture score (MFCS, max 3): curved lateral border**, **medial crease**, **lateral head of the talus coverage**,
  - **Hindfoot contracture score (HFCS, max 3): posterior crease**, **empty heel**, **rigid equinus**.
  - **HFCS over 2.5**: tenotomy likely needed.
- **Dimeglio classification** (**grades I-IV, 20-point scale**: soft-soft (benign) to stiff-stiff (very severe)).
- **Imaging:** not routinely needed (**ultrasound** occasionally); radiographs: **talocalcaneal parallelism** (**reduced Kite angle on AP: under 20 degrees** (normal 20-40 degrees); **lateral talocalcaneal angle under 25 degrees**).

4. Ponseti method (gold standard; about 95% initial correction)

- **Start early (ideally within the first 1-2 weeks of life)**; effective up to about 2 years (and even in older children with modifications).
- **Serial manipulation and long-leg (above-knee, knee flexed 90 degrees) plaster casts**, changed **weekly** (**about 5-7 casts**).
- **Order of correction: "CAVE"**:
  - **Cavus first**: **supinate the forefoot (elevate the first ray)** to align the forefoot with the hindfoot (**never pronate**: pronation worsens cavus),
  - **Adductus and varus together**: **abduct the supinated foot with counter-pressure on the lateral head of the talus** (**fulcrum = lateral talar head, NOT the calcaneocuboid joint**: Kite's error, which locks the calcaneus and prevents its correction),
  - **Hindfoot varus corrects spontaneously** as the calcaneus abducts/rotates under the talus,
  - **Equinus last**: **abduct to about 60-70 degrees (thigh-foot angle)** before addressing equinus.
- **Percutaneous Achilles tenotomy** (**about 80-90% need it**): when **dorsiflexion under 10-15 degrees** after abduction is achieved; under local anaesthetic; **final cast for 3 weeks** in about **60-70 degrees abduction and 15-20 degrees dorsiflexion**; tendon regenerates.
- **Errors: pronating the foot** (iatrogenic **rocker-bottom foot**), **thumb on the calcaneocuboid joint (Kite)**, **dorsiflexing before correcting varus/adduction** (**rocker-bottom/midfoot break**), **short-leg casts** (slip), **external rotation of the foot in the cast (lateral talar head as fulcrum required)**.
- **Bracing (foot abduction orthosis: Denis Browne bar / Steenbeek / Mitchell boots): affected foot about 60-70 degrees external rotation (abduction), normal foot about 30-40 degrees**, **shoulder-**

width bar, **dorsiflexion bend about 10-15 degrees**; **23 hours/day for 3 months, then night and nap time until about 4-5 years**.

- **Non-compliance with bracing = the commonest cause of relapse** (**relapse about 10-30%**: up to 80% with poor compliance vs about 6% with good compliance).
- **Relapse: dynamic supination** (**overactive tibialis anterior: walking on the lateral border**), **recurrent equinus**, adduction.
  - **Repeat casting ± repeat tenotomy**.
  - **Tibialis anterior tendon transfer (TATT) to the lateral cuneiform (third cuneiform)** for **dynamic supination** in a **child over about 2.5-3 years with flexible deformity and an ossified lateral cuneiform**.
- **Atypical (complex) clubfoot** (**short, fat, stiff foot; deep plantar and posterior creases; short hyperextended first toe; equinus severe**): **modified Ponseti (Ponseti complex clubfoot protocol)**: **avoid excessive abduction**, **hyperflexion of the knee**, abduct only to about 40 degrees, use the index finger on the lateral talar head and both thumbs on the plantar metatarsal heads to correct cavus.

5. Surgical management (residual/neglected/syndromic)

- **Posteromedial release (PMR, Turco/McKay/Carroll; Cincinnati incision)**: **historical for primary treatment** (stiffness, overcorrection, scarring, **pain and arthritis in adulthood**); now for **resistant/syndromic** feet.
- **Bony procedures (older children):**
  - **Lateral column shortening (Evans calcaneocuboid fusion/Lichtblau resection)** (**4-8 years**),
  - **Medial column lengthening (cuneiform opening wedge) + cuboid closing-wedge osteotomy** (residual adductus),
  - **Calcaneal osteotomy (Dwyer lateral closing wedge)** for residual hindfoot varus,

  - **Talectomy** (**arthrogryposis, rigid recurrent clubfoot**),
  - **Triple arthrodesis** (**over about 10-12 years**: salvage),
  - **Ilizarov/gradual correction (Joshi's external stabilisation system: JESS; Indian contribution)** for neglected/recurrent clubfoot in older children.
- **Neglected clubfoot in older children/adults**: **Ponseti casting still useful** in walking children (good results even up to about 10 years); **JESS/Ilizarov**, **triple arthrodesis**.

Part B: Other paediatric foot deformities

6. Metatarsus adductus

- **Commonest foot deformity in infants** (about 1 per 1,000); **forefoot adduction with a normal hindfoot (neutral/valgus) and normal ankle dorsiflexion** (no equinus: **distinguishes from clubfoot**); **"bean-shaped foot"**; associated with **DDH (about 5-10%)**.
- **Bleck classification** (heel bisector line): **normal: through the 2nd-3rd toe web**, **mild: through the 3rd toe**, **moderate: between the 3rd and 4th toes**, **severe: between the 4th and 5th toes**.
- **Flexibility** (passive correction beyond neutral) determines treatment: **flexible: observation/stretching (about 90% resolve spontaneously)**; **stiff/rigid: serial casting (best under 6-8 months)**; **surgery rarely** (**over 4-5 years** with persistent rigid deformity: **medial cuneiform opening wedge + cuboid closing wedge osteotomy**). **Avoid tarsometatarsal capsulotomy (Heyman-Herndon: high failure, arthritis)**.
- **Skewfoot (serpentine/Z-foot): forefoot adduction + hindfoot valgus** (**talonavicular lateral subluxation**): often **iatrogenic** (overcorrection of metatarsus adductus/clubfoot by casting in valgus); **medial cuneiform opening wedge + calcaneal lengthening** if symptomatic.

7. Congenital vertical talus (CVT; "rocker-bottom foot")

- **Irreducible dorsal dislocation of the navicular on the talar neck**, **talus vertical (plantarflexed)**, **hindfoot equinus and valgus**, **forefoot dorsiflexed and abducted**; **rigid rocker-bottom foot** with a **"Persian slipper" appearance**; **palpable talar head on the plantar-medial sole**.
- **About 50% associated with neuromuscular/genetic disorders**: **arthrogryposis**, **myelomeningocele**, **trisomy 13/18**, **prune belly**, **HOXD10 mutation (familial)**; **check the spine, hips, and neurological status**.
- **Diagnostic radiograph: forced (maximal) plantarflexion lateral view**: **the talar axis line stays plantar to the first metatarsal axis (the talonavicular dislocation does not reduce)** (**in congenital oblique talus, it reduces**). **Lateral talar-first metatarsal angle** (Meary) markedly abnormal; **dorsiflexion lateral**: fixed equinus of the calcaneus.
- **Treatment:**
  - **Dobbs method ("reverse Ponseti")**: **serial casting (plantarflexion and inversion: forefoot plantarflexed and inverted with counter-pressure on the talar head)** followed by **percutaneous pinning of the talonavicular joint** (after reduction) and **percutaneous Achilles tenotomy**; **± anterior tibialis transfer to the talar neck**; bracing (**like Ponseti boots**). **Best results**.
  - **Open reduction (single-stage dorsal approach or two-stage) for failed/late cases**: release of the **tibialis anterior, EHL, EDL, peroneus tertius**, dorsal talonavicular capsule; **Achilles lengthening**; **talonavicular pinning**.
  - **Older children: talectomy, subtalar arthrodesis (Grice), triple arthrodesis**.

- **Congenital oblique talus**: **flexible**; **talonavicular joint reduces in plantarflexion**: **casting/observation**.

8. Calcaneovalgus foot

- **Postural: foot dorsiflexed, lying against the anterior tibia, hindfoot valgus**: **flexible; intrauterine positioning**; commoner in **first-born girls**; **resolves spontaneously or with stretching (within weeks-months)**.
- **Differential diagnoses: CVT** (rigid: radiograph), **posteromedial bowing of the tibia** (**calcaneovalgus foot + posteromedial tibial bowing**: **spontaneous improvement; leg-length discrepancy of about 3-4 cm at maturity: epiphysiodesis/lengthening**), **myelomeningocele (L5)** (weak plantarflexors).

9. Flexible flatfoot (pes planovalgus) in children

- **Physiological in infants and toddlers** (**fat pad, ligamentous laxity**; the arch develops by about **6-10 years**).
- **Flexible: arch reconstitutes on tiptoe (Jack/Hubscher toe-raise test), hindfoot inverts on tiptoe**; usually **painless and benign**: **no treatment** (orthoses do not change the natural history).
- **Tight Achilles/gastrocnemius (Silfverskiöld test)** with symptomatic flexible flatfoot: **stretching**; **calcaneal lengthening osteotomy (Evans)** + gastrocnemius recession for **symptomatic, failed conservative care**.
- **Rigid flatfoot: tarsal coalition**, **CVT**, **inflammatory arthritis**, **peroneal spastic flatfoot**, **accessory navicular**, **neuromuscular** (cerebral palsy).

10. Tarsal coalition

- **Failure of segmentation** (autosomal dominant with variable penetrance) of tarsal bones: **fibrous (syndesmosis), cartilaginous**

(synchondrosis) or bony (synostosis); **about 1-2% (probably higher)**; **bilateral in about 50%**.

- **Calcaneonavicular (CN): commonest (about 53%)**; **presents age 8-12 years** (ossifies earlier); **talocalcaneal (TC; middle facet): about 37%**, **presents age 12-16 years**.
- **Presentation: rigid painful flatfoot**, **recurrent ankle sprains**, **peroneal spasm ("peroneal spastic flatfoot")**, **limited subtalar motion**.
- **Radiographs:**
  - **CN coalition: "anteater nose sign" (elongated anterior process of the calcaneus) on lateral**, best seen on the **45-degree internal oblique view**,
  - **TC coalition: "C sign" (continuous line of the talar dome and the sustentaculum)**, **talar beaking (dorsal talar head: traction spur, not arthritis)**, **short talar neck**, best seen on the **Harris (axial) view**; **CT (gold standard; extent of coalition as a percentage of the posterior facet)**.
- **Treatment:**
  - **Non-operative (first): cast/boot immobilisation 4-6 weeks, orthoses, NSAIDs.**
  - **CN coalition: resection with interposition of the extensor digitorum brevis (EDB) or fat** (good results).
  - **TC coalition: resection with fat/FHL interposition if the coalition is under about 50% of the posterior facet area and no significant hindfoot valgus (over about 16-21 degrees) or arthritis**; **otherwise subtalar or triple arthrodesis**; **calcaneal lengthening osteotomy** with resection for valgus.

11. Accessory navicular

- **Accessory ossicle at the medial navicular within the tibialis posterior tendon** (**about 10-15% of people**; often bilateral);

Geist/Coughlin types: I (os tibiale externum: sesamoid in the tendon), **II (synchondrosis: commonest symptomatic)**, **III (cornuate navicular: fused)**.

- **Medial midfoot pain, prominence**, associated with flatfoot.
- **Treatment: shoe modification, cast, orthoses**; surgery: **excision of the ossicle ± advancement of the tibialis posterior (Kidner procedure)** or simple excision.

12. Cavus foot in children

- **High arch, hindfoot varus**; **two-thirds have an underlying neurological cause**: **Charcot-Marie-Tooth (HMSN type I: commonest; PMP22 duplication on chromosome 17)**, **spinal cord lesions (tethered cord, diastematomyelia, syringomyelia, spinal tumours)**, **Friedreich ataxia**, **polio**, **cerebral palsy**, residual clubfoot. **Unilateral cavus = spinal cord pathology until proved otherwise (MRI spine)**.
- **CMT mechanism: weak tibialis anterior vs strong peroneus longus** (**plantarflexed first ray: forefoot-driven cavus**) and **weak peroneus brevis vs strong tibialis posterior** (**hindfoot varus**); **intrinsic weakness (claw toes)**.
- **Coleman block test:** a block under the lateral foot (first ray allowed to drop): **hindfoot varus corrects = flexible hindfoot (forefoot-driven)**: **forefoot surgery (dorsiflexion osteotomy of the first metatarsal) + plantar fascia release ± peroneus longus to brevis transfer** suffices; **does not correct = rigid hindfoot**: add **calcaneal (Dwyer/sliding) osteotomy**; **triple arthrodesis** for severe rigid deformity (avoid if possible: adjacent joint arthritis).
- **Tendon transfers: peroneus longus to brevis (balance)**, **tibialis posterior to the dorsum (weak dorsiflexion)**, **Jones transfer (EHL to the first metatarsal neck + IPJ fusion) for clawed hallux**, **Hibbs (EDL to the lateral cuneiform/cuboid)**.

13. Toe deformities

- **Curly toes** (3rd-5th; **FDL/FDB tightness**: flexion and medial rotation): **usually resolve; flexor tenotomy** if persistent (**over 3-4 years**).
- **Overlapping fifth toe (digitus minimus varus)**: **dorsal and medial**; symptomatic: **Butler procedure (racquet incision, dorsal capsule release)**.
- **Congenital hallux varus**: **associated with polydactyly, short first metatarsal, longitudinal epiphyseal bracket (delta phalanx)**.
- **Polydactyly (postaxial commonest: fifth toe; African descent)**: excise the duplicated (usually the most lateral/medial) toe at about **9-12 months**.
- **Syndactyly** of toes: no treatment needed (cosmetic).
- **Juvenile/adolescent hallux valgus**: **metatarsus primus varus**, often **family history**, **DMAA increased**; **delay surgery until skeletal maturity** (**recurrence is high when physes are open**); **hemiepiphysiodesis of the lateral first metatarsal physis** (guided growth) in younger children.
- **Köhler disease** (**osteochondrosis of the navicular**, **boys aged 4-6**; **sclerotic, flattened navicular**; **self-limiting**: **short leg walking cast 6-8 weeks** shortens symptoms).
- **Freiberg infraction** (**osteochondrosis of the second metatarsal head**, **adolescent girls**; **Smillie stages I-V**; rest, stiff-soled shoes; **dorsal closing-wedge osteotomy** (Gauthier), debridement).
- **Sever disease** (**calcaneal apophysitis**, **ages 8-12**, **active children**: **heel pain with activity**, **squeeze test**: **self-limiting**: heel cups, stretching, activity modification).

Exam pearl: **Clubfoot = CAVE; corrected in that order. Ponseti: supinate to correct cavus; abduct with counter-pressure on the lateral talar head (not the CC joint: Kite's error); Achilles tenotomy in about 80-90%; braces 23 h/day for 3 months then nights until 4-5 years. Non-compliance with bracing = commonest cause of relapse; dynamic supination = TATT to the lateral cuneiform.**

Exam pearl: **Talar neck = most deformed bone; navicular medially subluxed; anterior tibial artery often absent. Pirani score (MFCS + HFCS, max 6). Pronation during Ponseti = rocker-bottom foot.**

Exam pearl: **CVT: rigid rocker-bottom; plantarflexion lateral view: talonavicular dislocation persists (oblique talus reduces); 50% syndromic; Dobbs reverse Ponseti + TN pinning + tenotomy. Metatarsus adductus: normal hindfoot, no equinus, mostly resolves. Skewfoot = adductus + hindfoot valgus.**

Exam pearl: **Tarsal coalition: CN (anteater nose, 8-12 years, 45-degree oblique) vs TC (C sign, talar beak, 12-16 years, CT); resect with EDB interposition (CN) or fat (TC under 50% facet). Unilateral cavus = spinal MRI; bilateral cavus = CMT; Coleman block test decides hindfoot surgery. Köhler navicular (self-limiting), Freiberg 2nd MT head, Sever calcaneal apophysitis.**

Paediatric fractures

1. Why children's fractures are different

- **Periosteum** is thick, strong and osteogenic: it limits displacement, acts as a hinge for reduction (keep the intact periosteal side as the hinge) and produces fast, abundant callus.
- **Bone is more porous and less brittle:** it fails in compression (**torus/buckle**), on the tension side only (**greenstick**), or bends without fracture (**plastic deformation**, most often ulna and fibula; can block reduction of the partner bone).
- **Remodelling** is greatest when the child is young, the fracture is **near a fast-growing physis**, and the deformity is **in the plane of motion of the adjacent joint**. **Rotational deformity does not remodel.** Intra-articular step-off does not remodel.
- **Overgrowth:** femoral shaft fractures in children aged about 2-10 overgrow by roughly 1-2 cm, which is why 1-2 cm of shortening (overlap) is accepted in a spica.
- **Ligaments are stronger than the physis:** an apparent "sprain" in a child is often a physeal injury (e.g. Salter-Harris I of the distal fibula).
- **Fastest-growing physes:** proximal humerus (about 80% of humeral growth) and distal radius (about 75-80% of radial growth) in the upper limb; **distal femur (about 70% of femoral length)** and proximal tibia (about 55-60% of tibial length) around the knee. Roughly 1 cm a year at the distal femur and 0.6 cm at the proximal tibia.

2. The physis and Salter-Harris

- **Zones (epiphysis to metaphysis):** reserve (resting), proliferative, **hypertrophic** (maturation, degeneration, **provisional calcification**). Fractures usually propagate through the hypertrophic zone, the weakest region; the germinal (reserve and proliferative) cells stay

with the epiphysis, which is why most physeal fractures grow normally.

- **Salter-Harris classification:**

Type	Fracture line	Notes
I	Through the physis only	Infants and young children; radiographs may be normal (look for widening); distal fibula, proximal humerus in neonates
II	Physis + metaphyseal fragment (**Thurston-Holland**)	**Commonest (about 75%)**; the periosteum is intact on the fragment side
III	Physis + epiphysis (intra-articular)	Needs anatomical reduction (e.g. juvenile Tillaux)
IV	Metaphysis + physis + epiphysis	Intra-articular and crosses the germinal layer; **ORIF**; high risk of a bar (e.g. medial malleolus, lateral condyle humerus)
V	Crush of the physis	Usually diagnosed retrospectively when growth arrest appears; worst prognosis
VI (Rang)	Perichondral ring injury	Peripheral bar, angular deformity (lawnmower injuries)

- **Peterson type I** (transverse metaphyseal fracture extending into the physis, distal radius) is worth knowing for FET-level questions.
- **Highest growth-disturbance risk: distal femur** (even SH I and II, because of the undulating physis), proximal tibia, distal tibia SH III/IV.
- **Physeal bar:** diagnose with **MRI (or CT) mapping**. Resect with interposition (fat, cement) if it involves **under 50%** of the physis and the child has **at least 2 years (about 2 cm) of growth left**. Otherwise complete the arrest (epiphysiodesis), deal with the other bone, and correct the angulation and length.

- **Park-Harris growth arrest lines** that converge on the physis indicate a bar; parallel lines mean symmetrical growth.

3. Elbow (the high-yield region)

Ossification centres (CRITOE)

- **C**apitellum about 1 year, **R**adial head 3-5, **I**nternal (medial) epicondyle 5-7, **T**rochlea 7-9, **O**lecranon 9-11, **E**xternal (lateral) epicondyle 11-13 (girls earlier).
- The **medial epicondyle always appears before the trochlea**: an ossified fragment in the joint before the trochlea ossifies is an incarcerated medial epicondyle.
- **Anterior humeral line** passes through the middle third of the capitellum; the **radiocapitellar line** passes through the capitellum in every view (if not, look for Monteggia).
- **Posterior fat pad sign** = effusion = occult fracture (supracondylar, lateral condyle, radial neck).

Supracondylar humerus fracture

- Commonest elbow fracture in children (peak 5-7 years), mechanism fall on the outstretched hand. **Extension type about 97%**, flexion type about 3%.
- **Gartland classification (with Wilkins modification):**

Type	Description	Treatment
I	Undisplaced; posterior fat pad only	Above-elbow cast 3 weeks
IIA	Displaced, posterior cortex intact (hinge), no rotation	Closed reduction; pin if anterior humeral line misses the capitellum
IIB	Hinged but rotated or translated	Closed reduction and pinning

Type	Description	Treatment
III	Completely displaced	**Closed reduction and percutaneous pinning**
IV (Leitch)	Multidirectionally unstable (periosteal hinge lost)	Pinning; often diagnosed under the image intensifier

- **Nerves: anterior interosseous nerve is the commonest** (extension type), then radial, median and ulnar.
  - **Posterolateral displacement** of the distal fragment (proximal spike goes anteromedial): **median nerve and brachial artery**.
  - **Posteromedial displacement** (spike goes anterolateral): **radial nerve**.
  - **Flexion type: ulnar nerve**.
  - **Iatrogenic:** ulnar nerve with a medial pin, especially when the elbow is hyperflexed (the nerve subluxes anteriorly).
  - Most nerve palsies are neurapraxias that recover in 2-3 months.
- **Pinning: 2 or 3 divergent lateral pins** separated at the fracture site are stable for most fractures with no ulnar nerve risk. Crossed pins are biomechanically stiffer; if a medial pin is needed, make a small incision and extend the elbow before placing it.
- **Vascular status:**
  - **Pink pulseless hand** (perfused, no palpable pulse) after reduction: admit, elevate and observe closely. Explore only if perfusion deteriorates or there is **associated median nerve palsy** (the artery may be tethered or entrapped).
  - **White pulseless hand** after closed reduction: **urgent open exploration** of the brachial artery (anterior approach).
  - **Compartment syndrome** (Volkmann ischaemia) risk is higher with ipsilateral forearm fracture (floating elbow) and with **casting in more than 90 degrees of flexion**.

- **Assessing reduction: Baumann angle** (shaft axis vs capitellar physis on AP; about 70-75 degrees, compare with the other side; change over 5 degrees = coronal malalignment), anterior humeral line, **teardrop**, medial and lateral column integrity.
- **Cubitus varus ("gunstock")** is the commonest late complication. It is due to **malunion (medial column collapse, internal rotation)**, not growth disturbance. It is cosmetic but can cause posterolateral rotatory instability, ulnar nerve symptoms and snapping triceps. Treat with a **lateral closing-wedge (French) osteotomy** or dome osteotomy.

Lateral condyle fracture

- Second commonest elbow fracture (age 5-10). It is **intra-articular and physeal (behaves like SH IV)**, pulled by the extensor muscles, so it displaces and has a high **non-union** risk.
- **Milch I** (lateral to the trochlear groove, SH IV, stable elbow) vs **Milch II** (into the trochlear groove, SH II-like, elbow unstable). **Jakob/Weiss** grade by displacement: under 2 mm, 2-4 mm with intact cartilage hinge, over 4 mm and rotated.
- **Internal oblique view** shows the true displacement best. An **arthrogram or MRI** shows whether the articular cartilage hinge is intact.
- **Treatment:** under 2 mm, cast with weekly radiographs (it can displace); **over 2 mm, ORIF** (posterior dissection is avoided: the blood supply enters posteriorly, so dissecting there causes **AVN of the condyle**).
- **Complications:** non-union, leading to **cubitus valgus** and **tardy ulnar nerve palsy** years later; lateral spur (overgrowth, cosmetic "pseudo-varus"); fishtail deformity.

Medial epicondyle fracture

- Ages 9-14; a traction (apophyseal) injury; about half are associated with **elbow dislocation**.

- **Absolute indication for surgery: incarceration of the fragment in the joint** (after dislocation reduction). Relative: ulnar nerve dysfunction, valgus instability in throwers or gymnasts, displacement in the dominant arm of an athlete.
- Most displaced fractures do well non-operatively (fibrous union is usually painless).

Radial neck, Monteggia and other forearm injuries

- **Radial neck:** angulation **under 30 degrees** is accepted; 30-60 degrees, closed reduction (Patterson manoeuvre, Israeli technique); over 60 degrees or translated, percutaneous joystick or **intramedullary (Metaizeau) reduction**. Avoid open reduction where possible: **loss of forearm rotation (stiffness) is the commonest complication**, and AVN of the radial head and radioulnar synostosis follow open reduction.
- **Monteggia (Bado):** I anterior radial head dislocation with apex-anterior ulnar fracture (**commonest in children**), II posterior, III lateral (often ulnar metaphyseal greenstick), IV anterior dislocation with both-bone fracture. In children, **restoring ulnar length and alignment usually reduces the radial head**; plastic deformation of the ulna must be corrected. A **missed Monteggia** (chronic) needs ulnar osteotomy with annular ligament reconstruction.
- **Galeazzi:** radial shaft fracture with distal radioulnar joint disruption; the paediatric equivalent is a distal ulnar physeal separation.
- **Forearm shaft:** acceptable alignment is generous in children under about 9 (angulation up to about 15 degrees, bayonet apposition, malrotation up to about 45 degrees); in older children less (about 10 degrees angulation, 30 degrees malrotation). Mid-shaft radius malunion limits rotation most. **Elastic stable intramedullary nails** for unstable fractures in older children. **Refracture** is common (about 5%) after greenstick fractures: complete the greenstick or cast long enough.

- **Distal radius:** the commonest paediatric fracture overall. Huge remodelling capacity: in a child with 2 years of growth left, up to about 20-25 degrees of sagittal angulation is acceptable. **Do not re-manipulate a physeal (SH II) fracture after 5-7 days** (physeal damage).
- **Nursemaid's elbow (pulled elbow):** annular ligament subluxation in children aged 1-4. Reduce by supination with flexion or by hyperpronation (more successful).

4. Proximal humerus and clavicle

- Proximal humerus physeal fractures (SH II in adolescents, SH I in younger children) remodel enormously (up to about 60 degrees accepted in young children). Operate only in adolescents with severe displacement or soft-tissue interposition (biceps tendon).
- **Neonatal clavicle fracture** (difficult delivery, pseudoparalysis): heals without treatment. Differentiate from **obstetric brachial plexus palsy** and **congenital pseudarthrosis of the clavicle** (right side, painless, no callus, subclavian artery pressure theory).
- **Medial clavicle physeal injury** (the last physis to close, about 23-25 years): posterior displacement can compress mediastinal structures. **CT**, then reduce in theatre with a thoracic surgeon available.

5. Spine in children

- **SCIWORA** (spinal cord injury without radiographic abnormality) is due to the elastic paediatric spine; **MRI** shows the lesion.
- Upper cervical injuries predominate under about 8 years (large head, horizontal facets). **Pseudosubluxation of C2 on C3** (up to about 4 mm) is normal; use the **Swischuk line**.
- Use a **back-board with an occipital recess or shoulder roll** in children under about 6 (the large occiput flexes the neck).
- **Chance (flexion-distraction) fracture** with a lap belt: look for associated **bowel injury**.

6. Pelvis and hip

- Pelvic fractures in children often need less fixation (elastic pelvis), but **associated visceral injuries** are frequent. The triradiate cartilage can arrest (shallow acetabulum).
- **Avulsion fractures in adolescent athletes:** ASIS (sartorius), **AIIS (rectus femoris)**, ischial tuberosity (**hamstrings**, the one that may need fixation if displaced over 1.5-2 cm), lesser trochanter (iliopsoas; **in an adult, a lesser trochanter avulsion = pathological fracture until proved otherwise**).
- **Delbet classification of paediatric hip fractures:**

Type	Site	Frequency	AVN risk
I	Transepiphyseal (with or without dislocation)	Rare	Highest, near 100% with dislocation
II	**Transcervical**	**Commonest (about 45-50%)**	About 50%
III	Cervicotrochanteric (basicervical)	About 30%	About 25%
IV	Intertrochanteric	About 10-15%	Lowest, about 10%

- Treat as an emergency: **urgent anatomical reduction**, stable fixation (avoid crossing the physis if possible in young children; stability is more important than sparing it), and **capsular decompression**. Complications: **AVN (commonest and most serious)**, coxa vara, non-union, premature physeal closure.

7. Femoral shaft

Age / size	Treatment
0-6 months	**Pavlik harness** or posterior splint

Age / size	Treatment
6 months-5 years	**Immediate (or early) hip spica**; accept up to about 2 cm shortening, about 15 degrees varus/valgus and about 20 degrees anterior/posterior angulation (less with age)
5-11 years	**Flexible (titanium) elastic nails** for length-stable fractures in children **under about 45-49 kg**; submuscular bridge plating or external fixation for length-unstable or comminuted fractures
Over about 11 years or heavier	**Rigid intramedullary nail through a lateral trochanteric entry**. Avoid piriformis entry: the lateral ascending cervical branches of the **medial femoral circumflex artery** pass there, so it risks **AVN of the femoral head**

- **Elastic nails:** two nails, each about **40% of the medullary canal diameter** (combined about 80%), pre-bent to the same curve so they balance. **Complications:** loss of length in unstable patterns, nail-end irritation, malunion in heavier children.
- A **femoral fracture in a child who is not yet walking** (under about 1 year) is a red flag for **non-accidental injury**.
- **Distal femoral physeal fractures** (often SH II) have a high rate of growth arrest (about 40-50%) even when well treated: warn the family and follow up for 1-2 years.

8. Knee

- **Tibial spine (eminence) fracture:** ages 8-14, the paediatric ACL equivalent. **Meyers-McKeever:** I undisplaced, II anterior hinge, III completely displaced (IIIA displaced, IIIB rotated), IV comminuted (Zaricznyj). The **intermeniscal ligament or anterior horn of the medial meniscus** can block reduction. Type I and reducible II: long-leg cast in extension. Irreducible II, III and IV: **arthroscopic or open reduction and fixation** (suture or screw).

- **Tibial tubercle avulsion:** adolescent boys jumping; Ogden classification. Watch for **compartment syndrome** (anterior tibial recurrent artery).
- **Patellar sleeve fracture:** a cartilage sleeve pulled off the inferior pole (ages 8-12); the radiograph shows only a small fleck of bone and patella alta, so the injury is underestimated. ORIF.
- **Proximal tibial metaphyseal fracture (Cozen):** a seemingly innocent undisplaced fracture in a child aged 3-6 develops **progressive genu valgum** over the next year from asymmetric overgrowth. Warn the parents; most remodel; avoid early osteotomy.
- **Proximal tibial physeal (SH) fractures:** risk of **popliteal artery injury** (the artery is tethered at the trifurcation).

9. Tibia and ankle

- **Toddler's fracture:** undisplaced spiral fracture of the distal tibia in a child just walking (9 months-3 years). May only show on an oblique view or at 10-14 days. Usually accidental.
- **Transitional fractures** occur in adolescents, because the distal tibial physis closes over about 18 months: **central first, then medial, then lateral**.
  - **Juvenile Tillaux:** SH III avulsion of the **anterolateral epiphysis** by the **anterior inferior tibiofibular ligament**, as only the lateral physis is still open.
  - **Triplane:** SH IV in three planes. It looks like SH III on the AP view and SH II on the lateral view; two-, three- or four-part types.
  - **CT** for both. **Articular displacement over 2 mm** needs reduction and fixation (screws parallel to the joint, avoiding the physis if growth remains).
- **Distal tibial SH III/IV (medial malleolus):** high risk of physeal bar and **varus** deformity. Anatomical ORIF.

10. Non-accidental injury (NAI)

- Red flags: a history inconsistent with the injury or the child's development, delay in presentation, injuries of different ages, and **any fracture in a non-ambulant infant**.
- **High-specificity fractures: classic metaphyseal lesions** (corner or bucket-handle fractures, from shaking or twisting), **posterior rib fractures** (squeezing), scapula, spinous process and sternum fractures.
- Moderate specificity: multiple or bilateral fractures, fractures of different ages, epiphyseal separations, vertebral body fractures, digital fractures in non-ambulant children, complex skull fractures.
- **Commonest fracture in abuse overall: diaphyseal long-bone fracture** (humerus or femur in infants); **humeral shaft fracture under 3 years** is suspicious.
- **Workup:** full **skeletal survey** in children under 2 (repeat at about 2 weeks), ophthalmology (retinal haemorrhages), head imaging, bloods (bone profile, vitamin D, coagulation). Differentials: **osteogenesis imperfecta**, rickets, scurvy, Caffey disease, congenital syphilis, normal metaphyseal variants.
- The treating doctor has a duty to **report concerns** (in India: child protection services; POCSO where sexual abuse is suspected) and to admit the child for safety.

> Exam pearl: **AIN is the commonest nerve injured in a supracondylar fracture; the ulnar nerve is the commonest iatrogenic injury (medial pin).** A **lateral condyle fracture over 2 mm displaced needs ORIF**, and its late complication is **tardy ulnar palsy from cubitus valgus**.

Exam pearl: **Delbet II is the commonest paediatric hip fracture; Delbet I has the highest AVN risk. Piriformis-entry nails are avoided in children because of AVN.**

Cerebral palsy and neuromuscular

1. Cerebral palsy: definition and classification

- A group of permanent disorders of movement and posture caused by a **non-progressive lesion of the immature brain** (before, during or within about 2 years after birth). The brain lesion is static, but the **musculoskeletal problems are progressive** (muscles do not grow at the rate of bone under spasticity, so contractures and bony deformity develop with growth).
- **Risk factors:** prematurity and low birth weight (strongest), perinatal hypoxia, intrauterine infection, multiple pregnancy, kernicterus (athetoid CP), neonatal stroke.
- **Physiological classification:**
  - **Spastic** (about 70-80%; upper motor neurone, velocity-dependent tone; pyramidal tract).
  - **Dyskinetic** (athetosis, dystonia, chorea; basal ganglia; kernicterus). **Poor candidates for tendon surgery** (unpredictable results).
  - **Ataxic** (cerebellum).
  - Mixed.
- **Topographic classification: hemiplegia** (arm worse than leg; walks by about 18-24 months; seizures common), **diplegia** (legs much worse than arms; classic for prematurity with **periventricular leukomalacia**), **quadriplegia / total body involvement** (all limbs, poor head control, intellectual disability, seizures, feeding problems).
- **GMFCS (Gross Motor Function Classification System), age 6-12 descriptors:**

Level	Function
I	Walks without limitation; difficulty only with advanced skills

Level	Function
II	Walks with limitations (long distances, uneven ground, stairs need a rail)
III	Walks with a hand-held mobility device (walker, crutches); wheelchair for long distances
IV	Self-mobility limited; powered mobility; may walk short distances with assistance
V	Transported in a manual wheelchair; very limited head and trunk control

- GMFCS is stable over time and predicts **hip displacement risk** (about 0% at level I to about 90% at level V) and walking prognosis.
- **Walking prognosis:** sitting independently by **age 2** predicts walking; not sitting by 4 suggests the child will not walk. Persistent primitive reflexes after age 1 (Bleck's criteria) are poor signs.

2. Principles of management

- **Team approach:** physiotherapy, orthoses, tone management, orthopaedic surgery. Goals depend on GMFCS: improve gait in I-III; keep sitting comfortable, hips painless, and spine balanced in IV-V.
- **Tone management:**
  - **Botulinum toxin A** (blocks acetylcholine release at the neuromuscular junction) for focal **dynamic** spasticity in young children (typically 2-6), effect lasts about 3-6 months. Combine with serial casting. Not useful for fixed contracture.
  - **Oral baclofen** (GABA-B agonist; sedation limits use); **intrathecal baclofen pump** for severe generalised spasticity or dystonia (GMFCS IV-V).
  - **Selective dorsal rhizotomy (SDR):** sectioning a proportion (often about half) of the L2-S1 dorsal rootlets, guided by intraoperative stimulation. Ideal candidate: **spastic diplegia, GMFCS II-III, age**

about 4-8, good strength and selective motor control, minimal fixed contractures, no dystonia. Permanent reduction of spasticity; may unmask weakness; later hip subluxation and spinal deformity can occur.

- **Gait analysis** (3D kinematics, kinetics, EMG) before multilevel surgery.
- **Single-event multilevel surgery (SEMLS):** correct all soft-tissue and bony deformities in one operation (rather than "birthday syndrome" of one operation a year), typically at **age about 6-10** when gait has matured. Followed by intensive rehabilitation for about a year.

3. The hip in CP

- **Spastic hip displacement** is due to overactive **adductors and iliopsoas** with weak abductors and extensors, leading to **coxa valga, increased femoral anteversion, and acetabular dysplasia (posterosuperior)**. The hip subluxes **posterosuperiorly** (unlike DDH).
- **Hip surveillance:** clinical exam and AP pelvis radiograph from about age 2, every 6-12 months, more often for GMFCS III-V (Australian/CPUP-based protocols).
- **Reimers migration percentage (MP):** percentage of the femoral head width lateral to Perkin's line. Normal under about 10%; **over 30-40% = subluxation**; 100% = dislocation. **Progression over about 10% a year** is significant.
- **Management by severity:**

Situation	Surgery
Young child (under about 6-8), **MP 30-40%** (some use 25-60%)	**Preventive soft-tissue release:** adductor longus and gracilis tenotomy ± iliopsoas lengthening (psoas over brim)

Situation	Surgery
MP over 40-50%, older child, or failed release	**Reconstructive: varus derotation osteotomy (VDRO)** of the proximal femur ± **pelvic osteotomy (Dega or San Diego)** for acetabular dysplasia, with open reduction if needed
Painful, long-standing dislocation with arthritic head in a non-walker	**Salvage:** proximal femoral resection (Castle or McHale), valgus osteotomy (Schanz), arthrodesis or arthroplasty

- Do not do an isolated obturator neurectomy (anterior branch) routinely; abandoned by most because of over-abduction (windswept or abduction contracture).
- **Windswept hips:** one hip abducted, the other adducted and dislocated, with pelvic obliquity and scoliosis.

4. The knee in CP

- **Crouch gait** (excessive knee flexion in stance): due to hamstring contracture, weak soleus or plantarflexor-knee-extension couple insufficiency (often **after isolated over-lengthening of the Achilles tendon**), lever-arm dysfunction (external tibial torsion, foot instability). Treat with hamstring lengthening, correction of lever arms, **distal femoral extension osteotomy** with **patellar tendon advancement** in severe fixed crouch, and floor-reaction AFOs.
- **Stiff-knee gait:** reduced and delayed knee flexion in swing from an **overactive rectus femoris** (confirmed by dynamic EMG; Duncan-Ely test). Treat with **distal rectus femoris transfer** (to the semitendinosus or gracilis) rather than release.
- **Hamstring tightness:** popliteal angle increased. Medial hamstring lengthening; lateral lengthening only if needed (risk of recurvatum and crouch). **Hamstring lengthening increases anterior pelvic tilt** (the hamstrings are hip extensors too).

- **Jump gait** (equinus + knee and hip flexion) and **true equinus** (hemiplegia). The Rodda and Graham sagittal gait classification: true equinus, jump, apparent equinus, crouch.

5. The foot in CP

- **Equinus** is the commonest deformity.
  - **Silfverskiold test:** if dorsiflexion improves with the knee flexed, the **gastrocnemius alone** is tight: **gastrocnemius recession** (Strayer, Vulpius, Baker). If not, both are tight: **tendo-Achilles lengthening** (Hoke triple hemisection, Z-lengthening).
  - **Avoid over-lengthening the Achilles tendon** in diplegia: it causes **crouch gait**, which is far worse than mild equinus. Diplegic equinus is often "apparent" (from knee and hip flexion).
- **Equinovarus** (hemiplegia): overactive **tibialis posterior** (split posterior tibial tendon transfer, SPLATT-type to peroneus brevis, or lengthening) or **tibialis anterior** (split anterior tibial tendon transfer, **SPLATT**). Confirm the culprit with dynamic EMG.
- **Equinovalgus / planovalgus** (diplegia and quadriplegia): peroneal overactivity and gastrocnemius-soleus contracture. **Calcaneal lengthening osteotomy (Evans)** in mobile feet; **subtalar or triple arthrodesis** in rigid or severe feet in older children.
- **Hallux valgus** develops with planovalgus: first MTP arthrodesis.

6. Spine and upper limb in CP

- **Neuromuscular scoliosis:** long **C-shaped thoracolumbar curves** with **pelvic obliquity**, mainly in GMFCS IV-V. Progresses even after skeletal maturity (especially curves over about 40 degrees).
- **Posterior spinal fusion from T2 to the pelvis** (pelvic fixation with iliac or **S2-alar-iliac** screws) when the curve exceeds about 40-50 degrees and sitting balance worsens. Complications are higher than in idiopathic scoliosis: infection, blood loss, pulmonary problems. Optimise nutrition first.

- **Upper limb (hemiplegia):** shoulder internal rotation, elbow flexion, forearm pronation, wrist and finger flexion, **thumb-in-palm**. Surgery (e.g. **flexor carpi ulnaris to ECRB transfer**, pronator teres rerouting, thumb adductor release) mainly improves function and hygiene; needs good sensation and voluntary control (the House and Zancolli classifications).

7. Muscular dystrophies

Duchenne muscular dystrophy (DMD)

- **X-linked recessive**, absent **dystrophin** (Xp21; deletions in about 65%); one-third are new mutations. Boys; incidence about 1 in 3,500-5,000 male births.
- Presents at **3-5 years**: clumsiness, frequent falls, **toe-walking**, **Gowers sign** (climbing up the thighs to stand: proximal weakness), **calf pseudohypertrophy** (fat and fibrous replacement), waddling gait, lumbar lordosis.
- **CK massively raised** (10-100 times normal, even at birth); diagnosis by **genetic testing** (multiplex ligation-dependent probe amplification); muscle biopsy (absent dystrophin on immunostaining) if needed.
- **Natural history:** loss of walking around **age 12** (later on steroids), scoliosis after loss of ambulation, death in the 20s-30s from **cardiomyopathy** or respiratory failure.
- **Corticosteroids** (prednisolone or deflazacort) prolong walking by about 2-3 years, reduce scoliosis and improve cardiorespiratory function. Exon-skipping drugs (e.g. eteplirsen for exon 51) and gene therapy for selected mutations.
- **Orthopaedics:** keep walking as long as possible (Achilles and ITB releases with KAFOs in selected cases). **Scoliosis:** fuse **early**, when the curve is **about 20-30 degrees**, before the **FVC falls below about 30-35%** (anaesthetic risk rises sharply). Long fusion to the pelvis or

L5. **Malignant hyperthermia-like reactions** and rhabdomyolysis: avoid succinylcholine and volatile agents.

- Fractures from osteoporosis (steroids, immobility): treat quickly to keep walking.

Becker and other dystrophies

- **Becker:** reduced or abnormal dystrophin; later onset (over about 7), walks beyond 16, milder, cardiomyopathy common.
- **Limb-girdle** (autosomal), **facioscapulohumeral** (AD; scapular winging, facial weakness: **scapulothoracic fusion** for winging), **Emery-Dreifuss** (early **elbow flexion contractures**, Achilles contracture, **cardiac conduction block**: pacemaker).
- **Myotonic dystrophy** (CTG repeat; distal weakness, myotonia, cataracts, frontal balding, cardiac conduction defects).

8. Spinal muscular atrophy (SMA)

- Autosomal recessive **SMN1 deletion**, loss of **anterior horn cells**: symmetric proximal weakness, absent reflexes, tongue fasciculation, **normal sensation and intelligence**. SMN2 copy number modifies severity.
- **Types:** I (Werdnig-Hoffmann; never sits, death in infancy without treatment), II (sits but never stands), III (Kugelberg-Welander; walks), IV adult.
- **Nusinersen** (antisense oligonucleotide, intrathecal), **onasemnogene abeparvovec** (gene therapy) and **risdiplam** (oral) have transformed survival.
- **Scoliosis** almost universal in type II: bracing for sitting, then **growth-friendly instrumentation** (growing rods, magnetically controlled) or fusion; leave access for intrathecal therapy if possible.
- **Hip subluxation and dislocation** are common but usually **painless: leave them alone**.

9. Myelomeningocele (spina bifida)

- Folic acid deficiency, valproate. Associated **Arnold-Chiari II malformation, hydrocephalus (shunt)**, tethered cord, syringomyelia, neurogenic bladder, **latex allergy** (latex-free theatre).
- **Motor level determines function:**

Level	Key muscle working	Typical function
Thoracic / L1-L2	Hip flexors only	Wheelchair; standing in a frame
L3-L4	**Quadriceps (L3-4), tibialis anterior (L4)**	Household or community walker with AFOs/crutches; **functional quadriceps is the key to walking**
L5	Hamstrings, toe extensors, gluteus medius partly	Community walker, AFOs
Sacral	Gastrocnemius-soleus	Near-normal walking

- **Hip dislocation** is commonest at L3-L4 (unopposed hip flexors and adductors vs weak abductors and extensors). **Do not reduce** a dislocated hip in most cases; correct contractures to keep the pelvis level.
- **Rapidly increasing scoliosis, new spasticity, deteriorating gait, changing foot deformity or urinary function:** suspect **tethered cord, shunt malfunction or syrinx**; MRI and neurosurgical review.
- **Feet:** aim for a **plantigrade, braceable foot**. Rigid clubfoot (talectomy in severe cases), vertical talus, calcaneus foot (L4-L5; anterior tibial transfer to the calcaneus). **Insensate feet** develop pressure ulcers and Charcot changes.
- **Fractures** in insensate limbs present as warm, swollen limbs with fever (mistaken for infection) and heal with exuberant callus.

10. Hereditary neuropathies and other conditions

- **Charcot-Marie-Tooth (HMSN):** commonest inherited neuropathy. **Type 1A: PMP22 duplication on chromosome 17**, demyelinating, slow conduction velocities. Type 2: axonal.
  - **Cavovarus foot**, claw toes, peroneal muscle wasting ("**inverted champagne bottle**" legs), intrinsic hand weakness.
  - **Mechanism:** weak **tibialis anterior** is overpowered by **peroneus longus** (plantarflexed first ray, forefoot-driven cavus); weak **peroneus brevis** is overpowered by **tibialis posterior** (hindfoot varus).
  - **Coleman block test:** a block under the lateral forefoot lets the first ray drop; if the **hindfoot varus corrects**, the hindfoot is flexible and the deformity is forefoot driven: plantar fascia release, **first metatarsal dorsiflexion osteotomy**, **peroneus longus to brevis transfer**, ± tibialis posterior transfer. If not, add a **lateralising calcaneal osteotomy**; triple arthrodesis only for rigid, arthritic feet.
- **Friedreich ataxia:** AR, **GAA trinucleotide repeat in frataxin** (chromosome 9). Ataxia from about age 10, areflexia, cavovarus feet, scoliosis (more like idiopathic in shape), **hypertrophic cardiomyopathy** (the main cause of death), diabetes.
- **Arthrogryposis multiplex congenita:** multiple **non-progressive** congenital contractures, normal sensation and usually normal intelligence. **Amyoplasia** (classic): internally rotated shoulders, extended elbows, flexed wrists, rigid **clubfeet**, dislocated hips, knee contractures. Early stretching; surgery aims for walking and independent function (bilateral hip dislocations are usually left; unilateral may be reduced).
- **Poliomyelitis (residual paralysis)** is still examined in India: asymmetric flaccid paralysis with normal sensation; tendon transfers once power has plateaued (about 2 years), "**a muscle loses one**

grade of power after transfer", and transfer **in phase** where possible.

> Exam pearl: **Silfverskiold test positive = gastrocnemius recession; over-lengthening the Achilles in a diplegic child causes crouch gait. Stiff-knee gait = distal rectus femoris transfer.**

> Exam pearl: **DMD scoliosis: fuse early (curve about 20-30 degrees, FVC above about 30-35%). SMA hip dislocation: leave alone. Myelomeningocele: L3-L4 (quadriceps) is the key level for walking.**

Skeletal dysplasias

1. Approach to a short child

- **Proportionate short stature:** constitutional, familial, endocrine (hypothyroidism, growth hormone deficiency), nutritional, systemic disease.
- **Disproportionate short stature = skeletal dysplasia** until proved otherwise:
  - **Short-limbed** (achondroplasia, pseudoachondroplasia, metaphyseal dysplasias) vs **short-trunk** (spondyloepiphyseal dysplasias, Morquio).
  - Pattern of limb shortening: **rhizomelic** (proximal: humerus and femur; achondroplasia), **mesomelic** (forearm and leg; Langer, dyschondrosteosis/Leri-Weill with **Madelung deformity**, SHOX gene), **acromelic** (hands and feet).
- Radiographic approach: which part of the bone is affected? **Epiphyseal** (MED, SED), **metaphyseal** (achondroplasia, metaphyseal chondrodysplasias, rickets), **diaphyseal** (Camurati-Engelmann), and is the **spine** involved (spondylo-)?
- **Upper cervical instability screen** (flexion-extension films ± MRI) before anaesthesia in **Morquio, SED congenita, pseudoachondroplasia, Down syndrome, Larsen syndrome, metatropic dysplasia**.

2. FGFR3 group

- **FGFR3 is a negative regulator of chondrocyte proliferation:** activating (gain-of-function) mutations slow growth at the physis.
- **Achondroplasia** (commonest skeletal dysplasia, about 1 in 25,000):
  - **Autosomal dominant**, **FGFR3 G380R** (Gly380Arg) in about 98%; **about 80% are new mutations** associated with **advanced paternal age**. Homozygous achondroplasia is lethal.

- **Clinical:** rhizomelic shortening, **frontal bossing**, midface hypoplasia, **trident hand** (separation of middle and ring fingers), **genu varum** (fibular overgrowth), thoracolumbar kyphosis in infancy (usually resolves when walking), lumbar hyperlordosis, normal trunk length, normal intelligence. Average adult height about 125-130 cm.
- **Radiographs: interpedicular distance narrows from L1 to L5** (normally widens), short pedicles, posterior vertebral scalloping, **champagne-glass pelvic inlet**, squared iliac wings, flat acetabular roofs, inverted-V physes at the knee.
- **Spine problems: foramen magnum stenosis** in infancy (central apnoea, hypotonia, **sudden death**; screen with MRI/polysomnography; suboccipital decompression); **thoracolumbar kyphosis** (avoid unsupported sitting; bracing; fusion if progressive over about 30 degrees after age 3-5); **lumbar spinal stenosis in adults** (neurogenic claudication; wide laminectomy and fusion).
- **Genu varum:** osteotomy or guided growth if symptomatic. Limb lengthening is controversial.
- **Vosoritide** (C-type natriuretic peptide analogue; antagonises FGFR3 signalling) increases annual growth velocity.

- **Hypochondroplasia** (FGFR3, milder; normal face). **Thanatophoric dysplasia** (FGFR3; lethal; "telephone-receiver" femora, cloverleaf skull).

3. COMP and type IX collagen group

- **Pseudoachondroplasia** (COMP, cartilage oligomeric matrix protein; AD):
  - **Normal face and normal length at birth**; short limbs become apparent at **2-3 years**.
  - Marked **ligamentous laxity**, windswept knees, **odontoid hypoplasia with C1-C2 instability**, small irregular epiphyses,

flared metaphyses, early **osteoarthritis** of hips and knees.
    - Distinguish from achondroplasia: normal face and skull, normal interpedicular distance.
- **Multiple epiphyseal dysplasia (MED)** (COMP, type IX collagen COL9A1-3, matrilin-3; usually AD):
    - Mild short stature, joint pain and stiffness, **small, irregular, flattened epiphyses**, **double-layered patella** (recessive type, SLC26A2).
    - **Bilateral, symmetrical femoral head changes at the same stage**, normal spine, acetabular involvement: differentiates it from **bilateral Perthes** (asynchronous, different stages).
    - **Premature osteoarthritis**: hip arthroplasty in early adulthood.

4. Type II collagen group (COL2A1)

- Type II collagen is in **articular cartilage, physis, vitreous humour and the inner ear**, which explains the eye and hearing problems.
- **Spondyloepiphyseal dysplasia (SED) congenita** (AD, COL2A1):
    - **Short-trunk** dwarfism present at birth, barrel chest, **coxa vara**, **odontoid hypoplasia / os odontoideum with atlantoaxial instability** (fuse C1-C2 if instability or myelopathy), kyphoscoliosis.
    - **Myopia and retinal detachment** (annual eye examination), hearing loss, cleft palate.
    - Delayed epiphyseal ossification (femoral heads may not be seen for years); early OA.
- **SED tarda:** X-linked (TRAPPC2/SEDL), boys, late childhood, hip and back pain, platyspondyly with a "heaped-up" posterior vertebral body.
- **Kniest dysplasia** (COL2A1): **dumbbell-shaped femora**, large stiff joints, myopia and **retinal detachment**, hearing loss, cleft palate.

- **Stickler syndrome** (COL2A1, COL11A1): myopia and retinal detachment, Pierre Robin sequence, early OA.

5. Other named dysplasias

- **Diastrophic dysplasia** (AR, **SLC26A2** sulfate transporter, common in Finland):
  - **"Hitchhiker" thumb** (short, proximally placed, abducted first metacarpal), **cauliflower ears** (cystic swelling in infancy), cleft palate.
  - **Rigid clubfeet** (very resistant to treatment), joint contractures, **cervical kyphosis** (can cause quadriplegia: monitor and fuse), scoliosis.
- **Cleidocranial dysplasia** (AD, **RUNX2/CBFA1**, master gene for osteoblast differentiation):
  - **Absent or hypoplastic clavicles** (shoulders can be brought together in front of the chest), **delayed closure of fontanelles** and wide sutures with **wormian bones**, **supernumerary teeth** and delayed tooth eruption, **coxa vara**, wide symphysis pubis.
  - Usually needs no treatment; coxa vara with Hilgenreiner-epiphyseal angle over 60 degrees needs valgus osteotomy.
- **Metaphyseal chondrodysplasias:**

Type	Gene / inheritance	Features
Schmid (commonest)	**COL10A1** (type X collagen), AD	Mild short stature, **coxa vara**, genu varum, waddling gait. **Looks like rickets but calcium, phosphate, ALP and vitamin D are normal**
Jansen	**PTH/PTHrP receptor** (activating), AD	Severe, **hypercalcaemia**, hypophosphataemia

Type	Gene / inheritance	Features
McKusick (cartilage-hair hypoplasia)	**RMRP**, AR (Amish)	Fine sparse hair, **immune deficiency**, Hirschsprung disease, risk of lymphoma
Shwachman-Diamond	SBDS, AR	**Pancreatic insufficiency**, neutropenia

- **Mucopolysaccharidoses** (lysosomal storage, defective glycosaminoglycan breakdown; urinary GAGs raised):

Type	Name	Inheritance	Key features
I	**Hurler** (alpha-L-iduronidase)	AR	Coarse face, **intellectual disability**, corneal clouding, gibbus (thoracolumbar kyphosis), carpal tunnel syndrome; stem-cell transplant
II	**Hunter** (iduronate sulfatase)	**X-linked recessive** (the only XLR MPS)	Like Hurler, **no corneal clouding**, milder
IV	**Morquio** (galactose-6-sulfatase; keratan sulfate)	AR	**Normal intelligence**, short trunk, pectus carinatum, **odontoid hypoplasia with atlantoaxial instability** (occipitocervical fusion often needed), ligament laxity, genu valgum, corneal clouding

  - **Radiographs (dysostosis multiplex):** oar-shaped ribs, bullet-shaped phalanges, anterior beaking of lumbar vertebrae (inferior beaking in Hurler, central in Morquio), proximally pointed metacarpals.
- **Osteopetrosis (marble bone disease): failure of osteoclast resorption**.

- **Infantile malignant** (AR; TCIRG1, CLCN7): marrow failure (anaemia, hepatosplenomegaly), cranial nerve compression (blindness, deafness), early death without **haematopoietic stem cell transplant** (osteoclasts are of haematopoietic origin).
- **Adult benign (Albers-Schonberg, AD, CLCN7):** fractures, osteomyelitis of the mandible.
- **Carbonic anhydrase II deficiency:** with **renal tubular acidosis** and cerebral calcification.
- **Radiographs:** dense, brittle bones, **bone-in-bone**, **Erlenmeyer flask** distal femora, **rugger-jersey spine**, sandwich vertebrae.
- Fractures heal slowly; **drilling and fixation are difficult** (very hard bone; use cooled drills and plates rather than nails).

- **Pyknodysostosis** (AR, **cathepsin K**): dense bones, short stature, **acro-osteolysis** of terminal phalanges, open fontanelles, obtuse mandibular angle, fractures. (Toulouse-Lautrec is thought to have had it.)
- **Camurati-Engelmann** (TGF-beta 1): symmetric **diaphyseal** cortical thickening, leg pain, waddling gait.
- **Melorheostosis:** "dripping candle-wax" cortical hyperostosis along one sclerotome; pain and contractures.
- **Osteopoikilosis** (multiple small sclerotic islands near joints, asymptomatic) and **osteopathia striata** (linear striations): incidental findings.

6. Osteogenesis imperfecta (OI)

- About **90% are COL1A1 or COL1A2 mutations** (type I collagen, quantity or quality), usually AD; recessive forms involve collagen-processing genes (CRTAP, LEPRE1/P3H1, SERPINF1 etc.).
- **Sillence classification:**

Type	Severity	Sclerae	Notes
I	Mild, non-deforming	**Blue**	Commonest; fractures decrease after puberty; **hearing loss** (third decade)
II	**Perinatal lethal**	Blue	Crumpled long bones, beaded ribs
III	**Most severe in survivors**, progressive deformity	Blue at birth, whiter with age	Very short, wheelchair-dependent, **dentinogenesis imperfecta**, triangular face
IV	Moderate	**Normal (white)**	Variable deformity
V	Moderate	Normal	**Hyperplastic callus**, **calcification of the interosseous membrane** of the forearm (limits rotation), radial head dislocation; IFITM5; normal collagen

- **Other features: wormian bones** in the skull, ligament laxity, easy bruising, basilar invagination, scoliosis (common, progressive; bracing ineffective; fuse curves over about 45-50 degrees), **dentinogenesis imperfecta**, hyperthermia under anaesthesia.
- **Bisphosphonates** (cyclical IV **pamidronate or zoledronate**) increase bone density and **reduce fracture rate and pain**; they do not cure the collagen defect. **Denosumab** has been tried (rebound hypercalcaemia on stopping).
- **Surgery:** correct long-bone bowing with **multiple osteotomies and intramedullary fixation** (**Sofield-Millar "shish-kebab" osteotomies**). **Telescoping rods** that elongate with growth (**Fassier-Duval**, Bailey-Dubow, Sheffield) reduce reoperation. **Avoid plates** (stress risers, peri-plate fractures).
- **Differential: non-accidental injury.** OI is suggested by blue sclerae, wormian bones, osteopenia, dentinogenesis imperfecta, family history, and fractures in hospital.

7. Neurofibromatosis type 1 (NF1)

- AD, **neurofibromin** (chromosome 17), a tumour suppressor (Ras-GTPase activating protein). Half are new mutations.
- **Diagnostic criteria (2 or more):** 6 or more **cafe-au-lait spots** (over 5 mm before puberty, over 15 mm after), 2 or more neurofibromas or one plexiform neurofibroma, **axillary or inguinal freckling**, optic glioma, 2 or more **Lisch nodules** (iris hamartomas), a distinctive osseous lesion (**sphenoid wing dysplasia or long-bone cortical thinning / tibial pseudarthrosis**), a first-degree relative with NF1.
- **Orthopaedic problems:**
  - **Dystrophic scoliosis: short, sharp, angular** curves (often kyphoscoliosis), **rib penciling**, vertebral scalloping, enlarged foramina, dural ectasia. **Rapid progression**; bracing fails; early **combined anterior and posterior fusion** (high pseudarthrosis rate). Non-dystrophic curves behave like idiopathic scoliosis but can "modulate" into dystrophic curves.
  - **Congenital pseudarthrosis of the tibia: anterolateral bowing** of the tibia in infancy (about half have NF1). **Protect with a total contact orthosis to prevent fracture**; avoid osteotomy for the bow. Once a pseudarthrosis develops: excision with **intramedullary rodding and bone grafting**, **Ilizarov bone transport**, or **vascularised fibular graft**; bone morphogenetic protein and cross-union techniques. Persistent non-union may end in amputation. (Compare: **posteromedial bowing** is benign and associated with calcaneovalgus foot and leg-length discrepancy; **anteromedial bowing** goes with fibular hemimelia.)
  - **Malignant peripheral nerve sheath tumour** (about 10% lifetime risk; a painful, enlarging plexiform neurofibroma needs MRI and PET).
  - **Hemihypertrophy**, overgrowth (elephantiasis neuromatosa), non-ossifying fibromas.

8. Tumour-like dysplasias

- **Hereditary multiple exostoses (multiple osteochondromas):** AD, **EXT1 (chromosome 8) or EXT2 (chromosome 11)**.
  - Sessile and pedunculated osteochondromas pointing away from the joint; short stature, **forearm deformity** (**ulnar shortening**, radial bowing, radial head dislocation), genu valgum, ankle valgus.
  - **Malignant transformation to secondary chondrosarcoma** in about 1-5% (much higher than a solitary osteochondroma, under 1%). Warning signs: **pain or growth after skeletal maturity**, **cartilage cap over 2 cm** in adults (MRI).
- **Ollier disease** (multiple **enchondromatosis**, unilateral predominance, non-hereditary; IDH1/IDH2 mutations): limb shortening and angular deformity; chondrosarcoma risk about 25-40%.
- **Maffucci syndrome** (enchondromas + **soft-tissue haemangiomas**, phleboliths): higher malignancy risk, including **non-skeletal malignancies** (gliomas, gastrointestinal cancers).
- **Fibrous dysplasia** (**GNAS1** activating mutation, postzygotic): polyostotic forms cause the **shepherd's crook** proximal femur. **McCune-Albright syndrome:** polyostotic fibrous dysplasia, **cafe-au-lait spots with "coast of Maine" borders**, precocious puberty.
- **Fibrodysplasia ossificans progressiva: ACVR1** mutation, **short great toes (hallux valgus) at birth**, progressive heterotopic ossification along fascial planes. **Never biopsy or operate** (flare-ups).

9. Chromosomal and connective tissue conditions with orthopaedic features

- **Down syndrome (trisomy 21): atlantoaxial instability** (atlanto-dens interval over about 5 mm), ligament laxity, **hip instability**

(dislocation in later childhood), **patellofemoral instability**, flatfoot, slipped epiphysis, scoliosis.

- **Marfan syndrome** (FBN1, fibrillin-1; AD): tall, arachnodactyly (**Steinberg thumb sign**, **Walker-Murdoch wrist sign**), **upward lens dislocation**, **aortic root dilatation**, scoliosis (larger, more rigid, higher complication rates), **protrusio acetabuli**, **dural ectasia**, pectus.
- **Homocystinuria** (cystathionine beta-synthase, AR): marfanoid habitus with **downward lens dislocation**, **intellectual disability**, osteoporosis and **thrombosis** (anaesthetic risk).
- **Ehlers-Danlos syndrome:** joint hypermobility, skin hyperextensibility; vascular type (COL3A1) has arterial rupture; avoid stabilisation surgery where possible (poor results).
- **Larsen syndrome** (filamin B): multiple congenital dislocations (knees, hips, elbows), cervical kyphosis, flat face.
- **Turner syndrome:** short stature, **cubitus valgus**, short fourth metacarpal, **Madelung deformity**.

Exam pearl: **FGFR3 = achondroplasia (interpedicular distance narrows caudally, foramen magnum stenosis). COMP = pseudoachondroplasia and MED. COL2A1 = SED congenita, Kniest, Stickler (eyes!). SLC26A2 = diastrophic (hitchhiker thumb). RUNX2 = cleidocranial. COL10A1 = Schmid. GNAS = fibrous dysplasia. EXT1/2 = HME. ACVR1 = FOP.**

Exam pearl: Screen the **C1-C2** level before anaesthesia in **Morquio, SED, pseudoachondroplasia and Down syndrome**; watch **cervical kyphosis** in diastrophic dysplasia and Larsen syndrome.

Bone and joint infection in children

1. Pathophysiology

- **Acute haematogenous osteomyelitis (AHO)** starts in the **metaphysis**: nutrient artery branches end in **sinusoidal loops beside the physis** where blood flow is slow and turbulent, the capillary endothelium has gaps, and there are few phagocytes. Minor trauma may create a haematoma that seeds bacteria.
- **Spread:** pus raises intraosseous pressure, tracks through the **Haversian and Volkmann canals** to the subperiosteal space, and lifts the loose paediatric periosteum (**subperiosteal abscess**). Cortical bone loses its blood supply (from both sides) and dies: **sequestrum**. The elevated periosteum lays down new bone: **involucrum**. Openings in the involucrum through which pus escapes: **cloacae**.
- **Age-related anatomy:**
  - **Under about 18 months: transphyseal vessels** cross the physis, so metaphyseal infection spreads to the **epiphysis and joint** (combined osteomyelitis and septic arthritis; physeal and epiphyseal damage).
  - **Childhood:** the physis is a barrier; infection stays metaphyseal and spreads subperiosteally (thick, loose periosteum).
  - **Adults:** periosteum firmly attached; infection spreads within the medullary canal and into the joint once the physes close.
- **Joints where the metaphysis is intra-articular** (osteomyelitis can rupture into the joint): **proximal femur (hip)**, **proximal humerus (shoulder)**, **proximal radius (elbow)**, **distal lateral tibia (ankle)**.
- **Septic arthritis destroys cartilage quickly:** bacterial toxins and **proteolytic enzymes from neutrophils** (proteoglycan loss within hours, collagen within days). The **hip is the emergency**: raised

intracapsular pressure can tamponade the retinacular vessels and cause **AVN of the femoral head**.

2. Organisms by age and setting

Setting	Organisms
Neonate (under 2 months)	**Group B streptococcus**, **Staphylococcus aureus**, Gram-negative enteric bacilli (E. coli), Candida in NICU babies
6 months-4 years	**S. aureus** and **Kingella kingae** (the commonest cause of osteoarticular infection in this age group where PCR is used)
Older children	**S. aureus** (by far commonest at all ages), group A streptococcus
Adolescents (sexually active)	Neisseria gonorrhoeae (septic arthritis, tenosynovitis, skin pustules)
Sickle cell disease	**Salmonella** is characteristic (bowel infarcts allow bacteraemia), S. aureus also common
Puncture through a sneaker	**Pseudomonas aeruginosa** (osteochondritis of the foot)
Unvaccinated child	**Haemophilus influenzae type b** (now rare)
Cat or dog bite	Pasteurella multocida
Chronic granulomatous disease	Aspergillus, Serratia, Burkholderia
India (endemic)	**Mycobacterium tuberculosis** in subacute cases, dactylitis, monoarthritis

- **MRSA** (community-acquired, especially **Panton-Valentine leukocidin (PVL)-positive** strains): severe sepsis, **multifocal**

disease, **deep vein thrombosis and septic pulmonary emboli**, necrotising myositis, more surgery.

- **Kingella kingae:** Gram-negative coccobacillus in the oropharynx of toddlers, often after a viral upper respiratory infection. **Mild illness, low-grade or no fever, normal or mildly raised CRP and WBC.** It grows poorly on standard media: **inoculate blood-culture bottles** with joint fluid and use **PCR**. Treat with a cephalosporin (resistant to clindamycin, glycopeptides).

3. Clinical features

- **AHO:** fever, malaise, **localised metaphyseal tenderness**, pseudoparalysis (refusal to use the limb), limp; swelling and erythema later.
- **Septic arthritis:** severely painful joint, **refusal to bear weight**, held in the position of maximum capsular volume (**hip: flexion, abduction and external rotation**), pain on any passive movement (log-roll), fever.
- **Neonates: few systemic signs**; irritability, pseudoparalysis, poor feeding; **multifocal** disease is common; high risk of joint destruction.
- **Psoas abscess, pyomyositis, discitis and sacroiliitis** can mimic septic hip: discitis in toddlers presents with refusal to walk or sit; pyomyositis (S. aureus, tropical) with a tender, woody muscle.

4. Investigations

- **Blood:** WBC, **ESR** (rises over days, falls slowly over weeks), **CRP** (rises within 6-8 hours, peaks at 48 hours, **normalises within about a week of effective treatment**: the best marker of response), **blood cultures** (positive in about 30-50%).
- **Kocher criteria (septic arthritis vs transient synovitis of the hip): non-weight-bearing**, **temperature over 38.5 C**, **ESR over 40 mm/h**, **WBC over 12,000/mm3**.

Criteria present	Predicted probability of septic arthritis
0	about 0.2%
1	about 3%
2	about 40%
3	about 93%
4	about 99.6%

    - **Caird** added **CRP over 2 mg/dL** (20 mg/L) as an independent predictor; **fever over 38.5 C is the best single predictor** in that study. Later validation studies showed lower accuracy in other populations: use the criteria as a guide, not a rule.

- **Imaging:**
    - **Plain radiographs:** soft-tissue swelling and loss of fat planes early; **bone changes (lysis, periosteal reaction) appear only after about 10-14 days** (30-50% loss of bone mineral needed). Baseline and to exclude differentials.
    - **Ultrasound: hip effusion** (absence of an effusion makes septic hip very unlikely), subperiosteal collections; guides aspiration.
    - **MRI** is the **investigation of choice for osteomyelitis**: marrow oedema (T1 low, T2/STIR high), subperiosteal and intraosseous abscess, physeal involvement, adjacent septic arthritis, pyomyositis. Contrast helps define abscesses.
    - Bone scan (Tc-99m) if the site is unclear or multifocal disease is suspected; can be cold early in severe infection (compromised blood flow).
- **Aspiration** (joint or subperiosteal pus) under ultrasound or fluoroscopy before antibiotics if it does not delay treatment: Gram stain, culture (plus blood-culture bottle and PCR for Kingella), cell count. **Synovial WBC over about 50,000/mm3 with over 75-90%**

neutrophils suggests septic arthritis (lower counts in Kingella and early infection).

5. Treatment

Antibiotics

- Start **empirical IV antibiotics immediately after cultures**: **anti-staphylococcal** cover (cefazolin or cloxacillin/flucloxacillin; **add vancomycin or clindamycin where MRSA is prevalent**; add cefotaxime/ceftriaxone in neonates and where Gram-negatives or Kingella are possible).
- **Short IV course with early oral switch** (current evidence, e.g. the Finnish trials of Peltola et al.): switch to high-dose oral antibiotics after about **2-4 days** once the child is clinically improving (afebrile, less pain) and **CRP is falling** (about 50% reduction).
- **Total duration: septic arthritis about 2-3 weeks**; **uncomplicated AHO about 3-4 weeks**; longer (6 weeks or more) for MRSA, complicated disease, neonates, pelvic and spinal infection, slow response.

Surgery

- **Septic arthritis of the hip is a surgical emergency: arthrotomy and thorough washout** (anterior approach; some use arthroscopy) as soon as possible. Other joints (knee, ankle, elbow) may be managed by arthroscopic washout or repeated aspiration in selected cases.
- **Osteomyelitis: operate** for a **subperiosteal or intraosseous abscess** (drill the cortex, drain the pus), **no clinical improvement after 48-72 hours** of appropriate antibiotics, sequestrum, or concomitant septic arthritis.
- Immobilise briefly for comfort (splint); early mobilisation once pain allows. Watch for **pathological fracture** in extensive osteomyelitis (protect with a cast).

6. Complications

- **Physeal arrest** leading to limb-length discrepancy or angular deformity.
- **AVN of the femoral head** and **hip dislocation or destruction** after septic hip (the neonatal destroyed hip is **Tom Smith arthritis**; Hunka and Choi classifications of sequelae).
- **Chronic osteomyelitis**, **pathological fracture**, recurrence.
- **DVT and septic emboli** (PVL-MRSA), sepsis and multi-organ failure.
- **Joint stiffness, early arthritis**, cartilage loss.

7. Subacute and chronic osteomyelitis

- **Subacute osteomyelitis (Brodie abscess):** low virulence or partial treatment; insidious pain, few systemic signs, often normal markers.
  - Radiograph: well-defined metaphyseal (or epiphyseal) lucency with a **sclerotic rim**, sometimes crossing the physis.
  - **Differentials: osteoid osteoma, Langerhans cell histiocytosis, chondroblastoma (epiphysis), Ewing sarcoma (diaphysis).**
  - **Roberts classification** of subacute osteomyelitis (types by location and appearance); biopsy and curettage when the diagnosis is uncertain.
- **Chronic osteomyelitis: sequestrum, involucrum, cloaca, sinus**.
  - **Treatment: sequestrectomy and debridement** to healthy bleeding bone (**paprika sign**), but **wait until the involucrum is strong enough** to support the limb (otherwise fracture or segmental loss); dead-space management (antibiotic beads or cement, bone graft later), soft-tissue cover, prolonged antibiotics guided by deep cultures.
  - Long-standing sinus: watch for **Marjolin ulcer (squamous cell carcinoma)** in adults.
  - **Cierny-Mader** staging (anatomical type I-IV and host A, B, C) is used in adults.

- **Chronic recurrent multifocal osteomyelitis (CRMO; chronic non-bacterial osteomyelitis):** autoinflammatory, **sterile cultures**, recurrent multifocal bone pain (**medial clavicle**, metaphyses, spine), associated **palmoplantar pustulosis, psoriasis or inflammatory bowel disease** (SAPHO in adults). Whole-body MRI; **NSAIDs** first line, then bisphosphonates or anti-TNF agents; not long-term antibiotics.

8. Special forms

- **Neonatal osteoarticular infection:** NICU babies with umbilical or central lines; multifocal; minimal signs; **combined osteomyelitis and septic arthritis** (transphyseal vessels); high rates of growth disturbance and hip destruction.
- **Discitis (toddlers, about 1-5 years):** the **disc is vascular in young children**. Refusal to walk or sit, back pain, abdominal pain. Mild fever, raised ESR/CRP. **MRI** shows disc space narrowing with endplate oedema. Treat with **antibiotics (anti-staphylococcal) and rest/brace**; biopsy rarely needed.
- **Pelvic osteomyelitis and sacroiliitis:** buttock, groin or abdominal pain, mimics septic hip or appendicitis; FABER test; MRI.
- **Puncture wound osteochondritis of the foot (Pseudomonas):** usually osteochondral (metatarsal heads, calcaneus); needs surgical debridement plus anti-pseudomonal antibiotics (ciprofloxacin, ceftazidime).
- **Sickle cell crisis vs osteomyelitis:** both cause bone pain, fever and raised markers. Infarction is far commoner; osteomyelitis is suggested by persistent fever, a single site, positive blood cultures, and fluid collections on ultrasound/MRI. **Salmonella cover** (e.g. ceftriaxone or ciprofloxacin) in empirical treatment. **Dactylitis (hand-foot syndrome)** in infants is infarction.
- **Gonococcal arthritis** (adolescents): migratory polyarthralgia, tenosynovitis, pustular skin lesions; ceftriaxone.

- **Lyme arthritis** (endemic regions): large-joint effusion with moderate pain and less fever; serology.
- **Osteoarticular tuberculosis in children (India):**
  - Insidious monoarthritis (hip, knee), **"spina ventosa"** (tuberculous dactylitis: expanded phalanx or metacarpal), spinal TB (see Spine notes).
  - **Phemister triad:** juxta-articular osteopenia, peripheral erosions, gradual joint space narrowing.
  - Diagnosis: histology (caseating granulomas), **GeneXpert/CBNAAT** (also detects rifampicin resistance), culture.
  - **Multidrug therapy** for about 12 months under the national programme; surgery for abscesses, debridement, deformity.

9. Differential diagnosis of the irritable hip in a child

Condition	Key features
Transient synovitis	Commonest cause of acute hip pain (3-8 years), often after a viral illness, mild or no fever, able to bear some weight, near-normal markers; resolves in 1-2 weeks with rest and NSAIDs
Septic arthritis	Toxic, febrile, refuses weight bearing, raised markers (Kocher, CRP)
Perthes disease	4-8 years, insidious limp, mild pain, normal markers, radiograph or MRI changes
SCFE	Adolescent, often obese, Drehmann sign
Osteomyelitis (proximal femur, pelvis)	Focal tenderness, MRI
Juvenile idiopathic arthritis	Morning stiffness, more chronic, other joints, uveitis risk

Condition	Key features
Malignancy (leukaemia, Ewing)	Night pain, systemic features, abnormal blood count, bone lesion

Exam pearl: **Metaphysis = commonest site; S. aureus = commonest organism at all ages; Kingella = 6 months-4 years (PCR, blood-culture bottles); Salmonella = sickle cell; Pseudomonas = sneaker puncture.**

Exam pearl: **MRI is the investigation of choice for osteomyelitis; radiographs lag 10-14 days. CRP is the best marker to monitor response. Septic hip = emergency arthrotomy.**